From measured attenuation to a wet-antenna law

Research log: what has been implemented, where we stand, and what comes next

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Data-driven discovery of rain-induced and wet-antenna attenuation laws in commercial microwave links

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Abstract

This report describes the research part of the project at the level of ideas: what each step does, why it is done that way, and what it found. It contains no code. The *baseline*, which turns the raw signal levels into the attenuation $A(t)$, is finished and frozen: it is unbiased, accurate to about $\pm 0.2\,\mathrm{dB}$ per event, and it does not absorb the drying of the antenna after rain. Stage 1, the wet-antenna attenuation (WAA), went through two phases. The first studied *drying tails*, the stretches of $A(t)$ after the rain has stopped. The tools were tested on simulated tails first, but on the real tails no drying law could be identified: many tails wait or rise before they dry, most of a tail passes while it still rains nearby, and the fast first drop does not grow with the path length. That phase was closed on 13 October 2026. The second phase treats each whole event as a system with the radar rain as input and $A(t)$ as output, $A = W + L\,P(R) + c$, where the antenna part W and the path part $L\,P$ are told apart only by the path length L, never by subtracting a power law. A ladder of models, each checked first on fake data, shows that WAA has memory: with a gauge as input it builds up in 2 to 3 minutes and dries in 9 to 13 minutes, on training and on validation events. A model-free test across all link lengths finds an antenna part that saturates at 2 to 3 dB and a path part that grows with the rain, so a constant WAA, the $f = L + L_0$ trap of the proposal, does not describe the data. The models fitted on all lengths pass this test on the training events but failed it on the validation events; dividing the radar rain by the event's radar-to-gauge ratio explains that gap. With that corrected rain the model was refitted and then tightened. The final model is a fast and a slow store with gauge timing; a more flexible slow store was dropped because its time constants could not be identified. The equation now has honest intervals from resampling whole events, and every number was checked on fake data. The path part is a straight line in the rain, $P = c\,R$, chosen from the data among physics forms and forms proposed by symbolic regression. The antenna level rises with the rain about like $\sqrt{R}$, but that closed form failed its check by a hair and the free curve is kept. The data fix the product of the antenna level and the link amplitude, not the level alone. Still to come are the validation of this WAA candidate without the test set, one reading of the test set, and then the path law $f(L, \bar{R}, Z)$ and the rain estimate.

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1 Scope and status at a glance

1.1 What this report covers

The report covers the research folder of the repository as it stood on 19 October 2026: the baseline module and the WAA module. The data layer below it (the pipeline that turns the raw OpenMRG files into a verified data set) has its own technical report and is summarised here in one page (Section 3). Every number in this report is taken from the documents the two modules keep: their method descriptions, their decision logs and their step reports (Appendix B). Unless stated otherwise, the numbers are for the *training* events and *training* links only. The validation events were first read in the verification step of Section 9, with models that had been frozen before; the test events and test links have not been touched.

The WAA steps carry two series of numbers. T0 to T18 are the drying-tail exploration (Sections 5 to 8; they were renamed on 13 October, and the old numbers such as 3b-1 or 4c are given in Table 1). S1 to S9 are the system identification that replaced it (Sections 9 and 10). Since the validation events have been used to make decisions, their numbers from S6 on are optimistic and are labelled so.

References in square brackets are those of the project proposal and are listed at the end.

1.2 Status

Figure 1 shows the whole project and Table 1 the steps inside it.

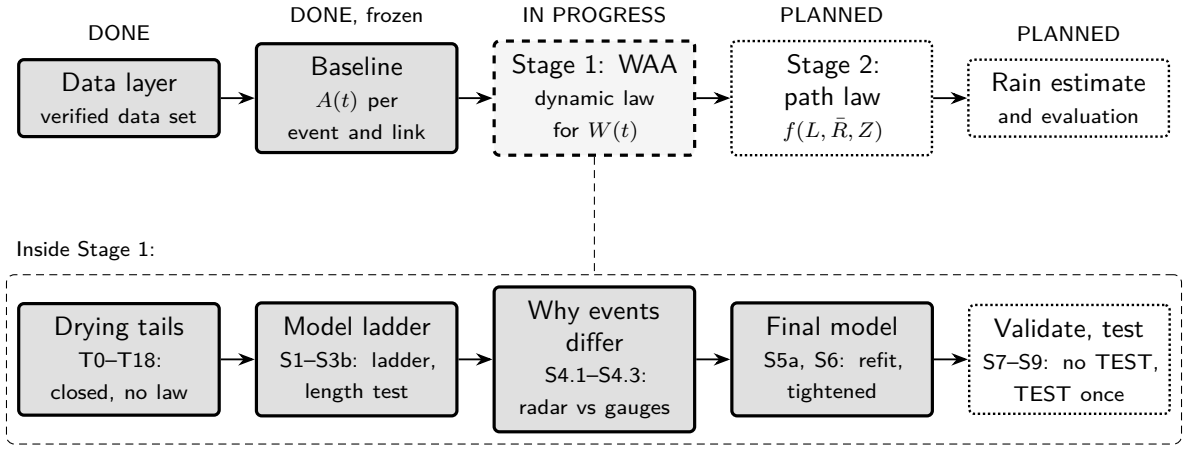


Figure 1: The project, and the inside of Stage 1. Thick solid border with grey fill: done. Thick dashed border: in progress. Dotted border: planned.

Table 1: Steps of the research part and their state on 19 October 2026. The old step numbers of the drying-tail exploration are given in brackets.

Step	What it is	State	Section
Baseline	Dry-weather level per (event, sub-link); $A(t)$	done, frozen 7 Oct	4
T0–T1 (0–1)	Inventory; extract every drying tail (20 355)	done	5
T2–T4 (2–2c)	Selection by fixed rules; frozen tail set v1 (429 train, 123 validation)	done, frozen 8 Oct	5
T5 (3a)	Simulator of tails, calibrated to the real set	done	6.3
T6 (3b-1)	Fit one tail at a time, on simulated tails	done: not enough	6.4
T7 (3b-1b)	Fit all tails together, on simulated tails	done: works	6.5
T8 (3b-1c)	Stress test of that fit	done: partly passed	6.6
T9 (3b-2)	SINDy directly on simulated tails	done: failed	6.7
T10–T11 (3b-3)	Two jumps: draw the drying curve, then name it	done: partly passed	6.8
T12–T13 (3b-3c, 3b-4)	The rest of the discovery method; a false step	done: partly passed	6.9
T14 (4)	Drying law $h(W)$ on the real tails, first fit	done: no law	7
T15–T18 (4b–7a)	What is in a drying tail; very short links	done; tails closed	8
S1	Fake check, then the model ladder on links below 1 km	done	9.2
S2, S2b	Links up to 3 km, a free rung, repairs; WAA v2	done	9.3
S3, S3b	Verification: model-free length test, validation events	done: VAL fails	9.4
S4.1–S4.3	Why events differ: radar against gauges	done	9.5
S5a	Refit on corrected rain; a flexible slow store; closed forms	done	10.1
S5b	Test set read once with the S5a model	not run (moved to S8)	10
S6	Tighten the equation: shared shapes, intervals, identifiability, closed forms	done	10.2
S7	Validate the WAA candidate without the test set	not started	12
S8, S9	Hashed freeze, test set once; write-up	not started	12
Old 5–9	Planned tail steps (drying conditions, wetting law, verification)	cancelled 13 Oct	8.5
Stage 2	Path law $f(L, \bar{R}, Z)$ with symbolic regression	not started	12
Evaluation	Rain estimate, comparison with the literature	not started	12

2 The problem in equations

2.1 What a link measures

A commercial microwave link (CML) sends a signal from one mast to another. The network records the transmitted signal level (TSL) and the received signal level (RSL). Their difference is the total loss $\mathrm{TL}(t) = \mathrm{TSL}(t) - \mathrm{RSL}(t)$. Most of it is always there (free-space loss, cables, antennas). What is always there is called the *baseline*, and the *attenuation* is what is left above

it:

$$A(t) \;=\; \mathrm{TSL}(t) - \mathrm{RSL}(t) - \mathrm{baseline} \;=\; A_{\mathrm{rain}}(t) + \delta(t). \tag{1}$$

A_{rain} is the loss caused by rain along the path. $\delta(t)$ collects everything else, mainly wet-antenna attenuation and whatever error is left in the baseline.

2.2 Rain along the path

Every short piece dx of the path attenuates the signal by $a\,R(x,t)^b\,dx$, where $R(x,t)$ is the local rain rate and a, b depend on the frequency and polarisation. The link sums all pieces:

$$A_{\mathrm{rain}}(t) \;=\; \int_0^L a\,R(x,t)^b\,dx \;\approx\; a\,\bar{R}(t)^b\,L, \tag{2}$$

with L the path length and $\bar{R}$ the mean rain rate along the path. The approximation is exact only when the rain is uniform along the path. The project replaces it by a more general form,

$$A_{\mathrm{rain}}(t) \;=\; a\,\bar{R}(t)^b\,f(L,\bar{R},Z), \tag{3}$$

where f is a correction function to be found from data and Z stands for properties of the path and the link (non-uniform rain, frequency, polarisation, time resolution). Uniform rain gives $f = L$.

2.3 Wet-antenna attenuation, and why it comes first

Water collects on the antenna covers during rain and dries afterwards. This adds an attenuation $W(t)$, the WAA. It differs from A_{rain} in two ways: it is local (it happens at the two antennas, so it does not grow with L), and it has memory (it stays after the rain has stopped).

WAA has to be removed before f is learned. Suppose the WAA were a constant W. Then

$$A = a\bar{R}^b L + W \quad\Longrightarrow\quad \frac{A}{a\bar{R}^b} = L + \underbrace{\frac{W}{a\bar{R}^b}}_{L_0}, \tag{4}$$

so a constant WAA looks exactly like a path that is longer by L_0. A method that learns f from A would return $f = L + L_0$ and report a length dependence that is not there. This fixes the order of the work (Figure 2): baseline, then WAA, then f.

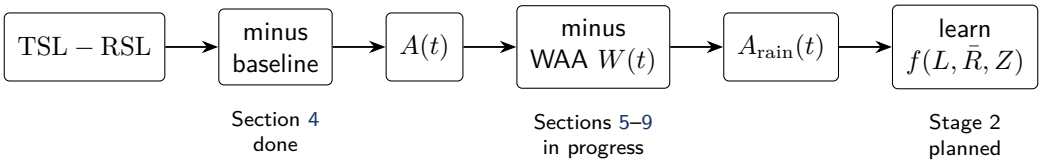


Figure 2: The order of the work. An error in an early block is passed on to every later block, which is why each block is validated and frozen before the next one starts.

2.4 Three rules that shape every step

1. **No circular selection.** Links or events are never chosen by how well a model fits them. Choosing the links where the power law works, and then concluding that the power law works, would prove nothing. Every selection rule in this report is fixed before the result is seen and never looks at the shape of the curve.

2. **Readable equations.** The aim is an explicit law that can be checked against the physics (a *gray-box* model), not only a good prediction. A neural network [11] is used later only as a reference for performance.
3. **Rain is defined by the references, never by the link.** Whether it rains on a link is decided from the weather radar along the path and from the rain gauges. The link's own signal is never used for that, because the signal is the thing being explained.

3 What the data layer provides

The research uses the OpenMRG data set [13]: three summer months (June to August 2015) of 364 two-way links (728 sub-links, one per direction) around Gothenburg, with ten city rain gauges, one national gauge with weather data, and a weather-radar composite. The points that matter for the research are these.

- **Resolution.** The links are sampled every 10 s. RSL is recorded in steps of 0.3 dB and TSL in steps of 1 dB. This *quantization* is the size of the effects studied here, so it appears in almost every section.
- **Rain reference.** The radar gives the rain rate every 5 minutes on a 2 km grid; the data layer averages it along each link path to get $\bar{R}$. The radar is preferred over a single gauge because it sees the whole path.
- **Rain events.** 89 rain events and 62 long dry periods are defined from the radar and the gauges only.
- **Reference state of each link.** Every 5 minutes each link is labelled *wet* (radar on the path or a gauge within 1 km shows at least 0.1 mm/h), *dry* (no rain at all on the path in that bin and for 30 minutes on either side), or *unknown* (anything in between, or radar missing). Unknown never counts as dry.
- **Split.** Events are divided into training, validation and test sets (60 / 20 / 20 %), and links into training and test (80 / 20 %). The split is frozen. The training set has 283 links (566 sub-links).

4 Step 1: the baseline and $A(t)$ (done, frozen)

4.1 The idea and the main risk

The baseline is the total loss a sub-link would show if it were not raining. It cannot be measured during rain, so it has to be estimated from dry time. The main risk is specific to this project. After the rain stops, the antenna is still wet and the signal returns slowly. If the baseline is estimated from that period, or is allowed to move during the event, it follows the slow return and the wet-antenna attenuation disappears into the baseline, which is exactly the signal Stage 1 needs. The baseline must therefore be blind to everything that happens after the event starts.

4.2 The method

For each pair of a rain event e and a sub-link s, the baseline is one number: the median of the total loss over the usable dry time in the 24 hours before the event starts,

$$\begin{aligned} B_{e,s} &= \operatorname{median}\{\, \mathrm{TL}_s(t) \;:\; t \in [\, t_{\mathrm{start}}(e) - 24\,\mathrm{h},\; t_{\mathrm{start}}(e)\,),\; t \text{ usable} \,\}, \\ A_s(t) &= \mathrm{TL}_s(t) - B_{e,s}. \end{aligned} \tag{5}$$

A 5-minute slot is *usable* if the link's reference state is dry and at least one hour has passed since the link was last wet (so that the drying of an earlier shower is kept out). If the 24-hour window holds less than one hour of usable time, a 48-hour window is used; if that fails too, the pair gets no baseline rather than a guess. This happens for 2.9 % of the pairs, almost all of them in the first events of the season, which have no history before them. Figure 3 shows the construction.

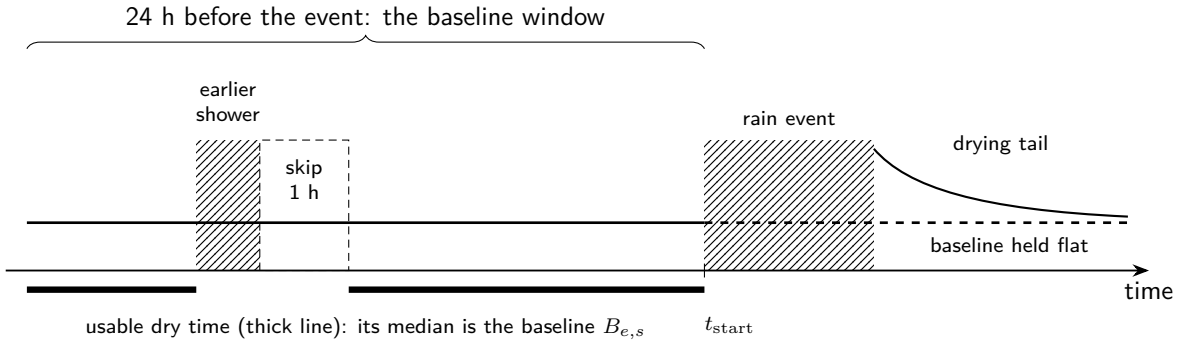


Figure 3: The baseline of one event on one sub-link. Only dry time before the event is used, so the baseline cannot follow the drying tail. Hatched: rain according to the references.

A first version of the method (v1) used the mean over 6 hours before the event, and in some cases a sloping line from the level before the event to the level after it. The frozen version (v2) differs in three choices, each one decided by the validation below: a longer window (24 h), the median instead of the mean, and a flat baseline only. The sloping line was dropped because the cases where it applied did not look like drying; they looked like changes of level that have nothing to do with the rain.

4.3 How a baseline is validated when the truth is unknown

There is no "true" baseline to compare with during rain. The idea of the validation is to create cases where the truth *is* known. In certified dry weather the attenuation is zero by definition. So we place *fake events* in dry periods: we pretend that an event of duration D starts at a random dry moment, compute the baseline exactly as for a real event, and measure what the method reports during the fake event. Anything other than zero is baseline error. The error of one fake event is its *offset*,

$$\text{offset} \;=\; \frac{1}{D}\int_{\text{fake event}} \big(\mathrm{TL}(t) - B\big)\, dt, \qquad \text{truth} = 0. \tag{6}$$

This was done for 513 good training sub-links, six durations from 1 to 40 hours and 20 fake events each: 61 560 fake events. Table 2 gives the result.

Table 2: Baseline error on fake events in dry weather (truth: zero). "90 % within" is the 90th percentile of |offset|. The summer median is the simplest possible baseline (one number per sub-link for the whole season).

	Duration D of the fake event					
90 % of offsets within (dB)	1 h	3 h	6 h	12 h	24 h	40 h
Summer median	0.43		0.40		0.34	
v1: 6 h mean before the event	0.29	0.31	0.34	0.36	0.30	0.38
v2: 24 h median before the event	0.33	0.32	0.31	0.28	0.28	0.29
v2: median offset	0.00	0.00	0.00	0.00	+0.02	+0.05

What the table says:

- The baseline is **unbiased**: the median offset is zero.
- Its **spread is about** ±0.2 dB (10th to 90th percentile), and 90 % of the errors are below about 0.3 dB. That is one RSL quantization step: the baseline is about as accurate as the instrument allows.
- The short window (v1) is slightly better for events of 1 to 3 hours, the long window (v2) is better from 6 hours on. Real significant events last several hours, so v2 was chosen.

4.4 Does the baseline absorb the drying?

This is the check the proposal asks for. After each real event we measure how far the signal is above the baseline k hours after the link's last wet moment, and compare with the same quantity after fake events (the control, where nothing dries). If the baseline had swallowed the drying, the real and the control curves would be equal at all k. Figure 4 shows a small excess after real rain (+0.04 dB above the control one hour after the rain) that shrinks to zero by 12 hours. The drying signal stays in $A(t)$, as intended. The excess is small in the median because most (event, sub-link) pairs have little rain; the strong cases are selected in Section 5.

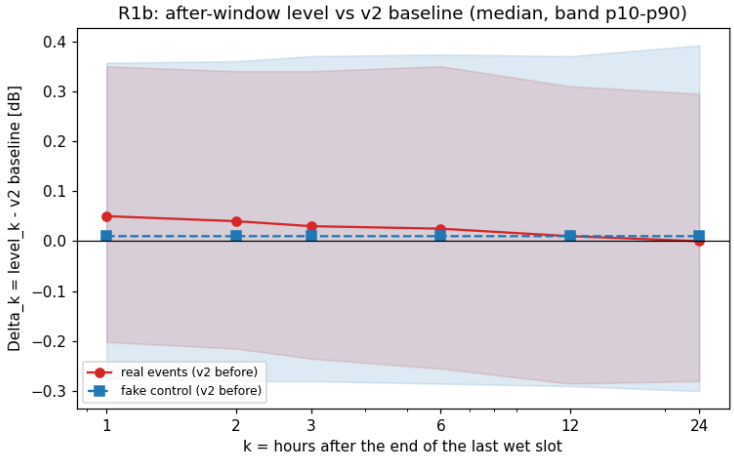


Figure 4: Level after the rain minus the baseline, against the time since the link's last wet moment. Circles, solid line: real events. Squares, dashed line: control (fake events in dry weather). Bands: 10th to 90th percentile.

4.5 What disturbs the signal in dry weather

The validation also showed what the baseline cannot remove, because it happens with no rain at all:

- **A daily cycle.** The dry-weather loss of a sub-link moves over the day, by 0.15 dB for the median sub-link but by more than 0.4 dB for 14 % of them. Those sub-links have about twice the baseline error.
- **Dew.** Jumps of 1 dB or more in dry weather happen mostly at night in humid air (a third of them at relative humidity ≥ 90 %, against 5 % of all dry time), usually in both directions of a link and on neighbouring links at once. This is most likely dew on the antennas: wet-antenna attenuation without rain. An hour is marked "at dew risk" when the humidity is at least 90 % or the air is within 2 °C of its dew point.

These are not corrected. Instead each (event, sub-link) pair carries *flags* (baseline far from the usual dry level, large daily cycle, dew risk, degraded link), and the pairs with a serious flag are left out of the main WAA analysis. One tested idea was rejected by a rule fixed before the run: leaving the dew-risk hours out of the baseline window did not improve the error, so it was not adopted.

4.6 What the baseline hands to the next step

- One baseline per (event, sub-link), for training and validation events on training links, with its flags. $A(t)$ follows from Eq. (5).
- One number per sub-link, σ_{off}: the spread of its baseline error on fake events. It says how far $A(t)$ of that sub-link may sit from zero for no physical reason, and it is used later as prior knowledge when laws are fitted (Section 6.5).

In short. The baseline is the median dry level of the 24 hours before each event, held flat. It is unbiased, accurate to about one quantization step, and blind to the drying after the rain. Frozen on 7 October 2026.

5 Stage 1, part 1: the drying tails (done, frozen)

5.1 The idea

During rain, $A(t)$ is the sum of two unknowns, $A_{\text{rain}}(t)$ and $W(t)$, and they cannot be separated without assuming a model for one of them. After the rain has stopped on a link, the path term is zero:

$$R(x,t) = 0 \text{ on the path} \implies A_{\text{rain}}(t) = 0 \implies A(t) = W(t) + \text{baseline error} + \text{noise}. \quad (7)$$

The stretch of $A(t)$ after the rain is called a *drying tail*. In it we observe the WAA almost directly, with no assumption about the power law. Figure 5 shows the picture. The tails show only how the antenna *dries*; how it *gets wet* has to be learned later from data during the rain (Section 12).

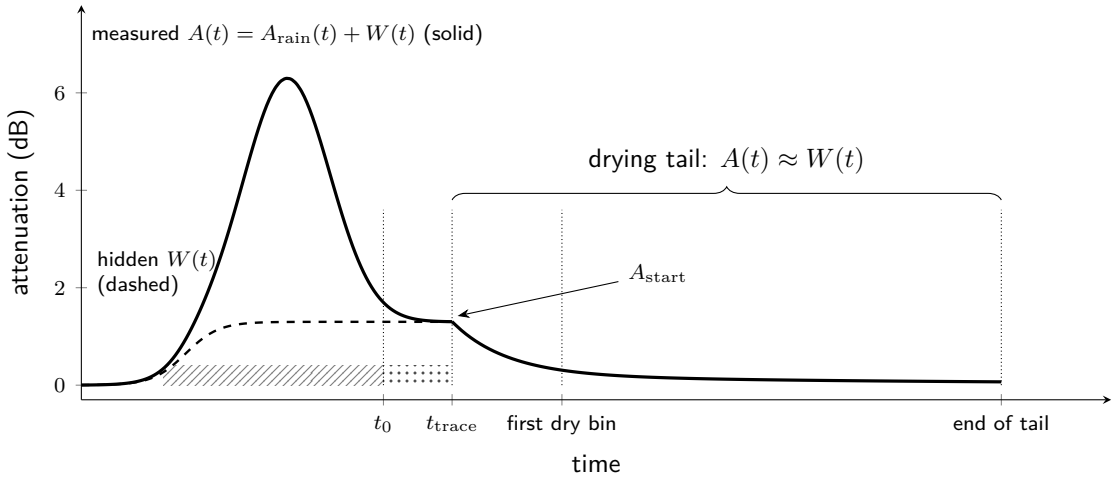


Figure 5: Sketch of one event on one link (not real data). During the rain the measured attenuation is the sum of the path term and the hidden WAA. After the last trace of rain, at t_{trace}, only the WAA is left and it decays as the antenna dries. Hatched bar: rain on the path according to the references. Dotted bar: only a trace of rain, below the threshold that makes a bin wet.

5.2 Extracting every tail, with no selection

The first step cuts one tail for every (event, sub-link) pair that has a baseline and had rain, using the reference state of the link only:

- **Start** t_0: the end of the link's last wet bin in the event.
- **End**: the first of (a) the next wet bin of the link, (b) a gap of 15 minutes or more in the data, (c) $t_0 + 12$ hours. The end never looks at the value of A, so a tail is not forced to reach zero.

This gives 20 355 tails on 566 sub-links from 69 events (15 370 from training events). Nothing is selected and nothing is fitted at this point. A first descriptive result already shows that the memory exists: of the tails that start above 2 dB, only 25 % are back at their dry level after 10 minutes and 79 % after 3 hours, while tails that start below 0.5 dB are back within 10 minutes in 94 % of the cases.

5.3 Selection by rules fixed in advance

Most of the 20 355 tails are too weak or too disturbed to say anything about drying. The selection uses only rules that are fixed before the result is seen and that never look at the shape of the decay:

- **Exclude** pairs with a flagged baseline, sub-links that are not of good quality, sub-links whose dry-weather baseline error σ_{off} exceeds 0.3 dB (one RSL step), and tails with more than 10 % of missing minutes.
- **Shorten** a tail when something other than drying starts: a step in the transmitted power, or a sustained rise of A (below).
- **Main set**: the tail starts at $A_{\text{start}} \geq 1$ dB and has at least 60 usable minutes. A larger *alternative set* uses $A_{\text{start}} \geq 0.6$ dB and serves as a sensitivity check.
- **Control set**: tails that start below 0.5 dB. They show what a tail looks like when there is nothing to dry, and are used to measure the noise (Section 6.3).

5.4 A lesson: the "fast drop"

The first selection produced 959 main tails, and they showed a striking feature: within the first half hour (the margin between the last wet bin and the first certified dry bin) the attenuation fell by a median of 1.1 dB, which is 83 % of its starting value. It was tempting to read this as a fast drying of the antenna. Two checks said otherwise.

- The drop did not grow with the link length L, which would point to the antenna and not to the path. But the comparison quantity, the attenuation during the rain, barely grew with L either (rank correlation 0.13), so this test had little power to decide.
- Looking at the references more closely showed that in this margin the radar or a gauge often still reported a *trace* of rain, below the threshold that makes a bin "wet". When the start of the tail was moved to the end of the last trace, 381 of the 959 tails no longer started above 1 dB.

So much of the fast drop was light rain ending, not the antenna drying. The consequence is the stricter start used from then on, and a clear statement of what is and is not shown: the tails are a clean place to study drying, but the proof that what dries is the antenna and not residual path rain is left to the test with very short links (Section 12), where the contrast in L is large enough.

5.5 The frozen tail set v1

The final rules are:

- **Start** t_{trace}: the end of the last 5-minute bin with any trace of rain (radar above zero on the path, a gauge within 1 km above zero, or radar missing, which counts as "unknown, so possibly rain"). A_{start} is the mean of A over the first 5 minutes from t_{trace}.
- **End**: the first of the ends listed above, the first new trace of rain, and the *rise cut*: if A (smoothed over 10 minutes) rises 0.6 dB above its lowest value so far and stays there for 10 minutes, the tail ends just before the rise began. A drying antenna does not get wetter, so a sustained rise means something else has started.

Figure 6 shows how many tails each stage leaves, and Table 3 the frozen set.

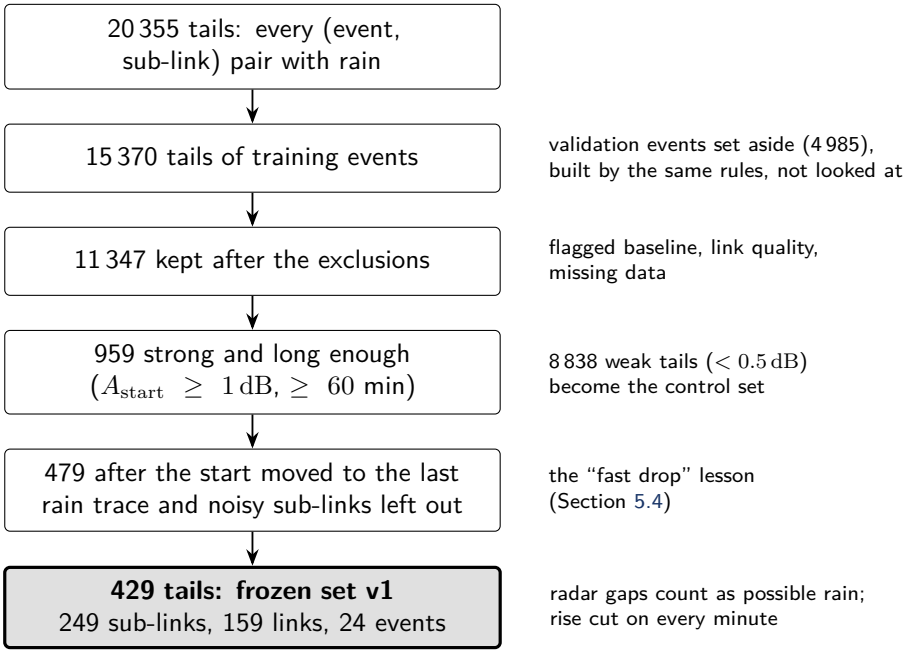


Figure 6: From all tails to the frozen set. No stage looks at how the tail decays.

Table 3: The frozen tail set v1 (8 October 2026).

Set	Tails	Sub-links	Links	Events
Main, training	429	249	159	24
Main, validation (not looked at)	123	96	76	9
Alternative ($A_{\text{start}} \geq 0.6$ dB), training	1 155	368	212	34
Alternative, validation (not looked at)	458	263	170	12

5.6 What the tails look like

Figure 7 shows the median of the 429 tails. It starts near 1.5 dB, loses half of its value in about 15 minutes, is at about a quarter after 30 minutes, and then slows down: about 14 % is left after one hour and 5 % after three hours. The control tails stay at zero. The shape is *fast at first, slow later*, and the central question of the next section is what kind of law produces it.

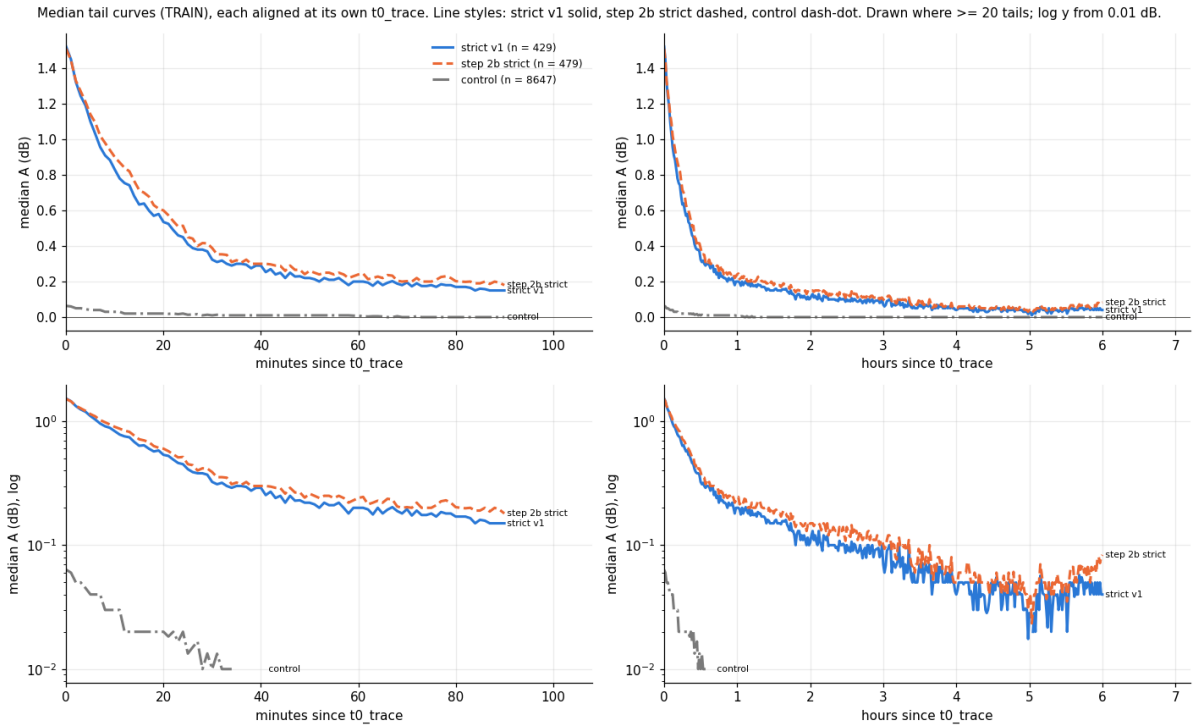


Figure 7: Median of the drying tails, each aligned at its own start. Solid: frozen set v1. Dashed: the set before the last two rules. Dash-dot: control tails. Top: linear scale; bottom: logarithmic scale, on which a single exponential would be a straight line.

Two properties of the set limit what can be concluded from it and are carried into every later step:

- **Three events hold 246 of the 429 tails.** Tails of one event share the same weather, so they are not independent. All uncertainties are therefore computed by resampling whole *events*, not single tails.
- **Some rises are unexplained.** In one night event the attenuation rises by 0.5 to 1 dB right after the rain with no trace in radar or gauges, then decays. It may be drizzle below the radar beam or condensation on the wet covers. The humidity record is missing for those hours, so the data cannot decide; the rise cut removes these stretches, and nine such events are flagged so that results can be recomputed without them.

In short. 429 drying tails, selected by rules that never look at the decay, are the frozen input of the WAA model. The median tail halves in about 15 minutes and then has a slow remainder lasting hours. Whether the fast early part is the antenna or the last light rain is not settled by the tails alone.

6 Stage 1, part 2: testing the tools on simulated tails (done for drying)

6.1 Why simulate

The law itself will be learned from real measurements only. But before a fitting method is trusted on real tails, we need to know what it can and cannot tell us. On real data that question has no answer, because the true law is unknown. On simulated data it has: we generate tails from a law we choose, pass them through the same distortions as the real measurements, apply the method,

and see whether it returns the law we put in. This is the same idea as the fake events of the baseline: build a case where the truth is known.

The simulated tails are not invented from scratch. Each one copies a real tail of the frozen set: the same sub-link, the same event, the same length, the same starting level, the same missing minutes. Only the decay and the noise are generated.

6.2 The candidate drying laws

Five laws are used, each a different physical picture of how the water leaves the antenna (Table 4, Figure 8). All start from $W(0) = A_{\text{start}}$ at the start of the tail.

Table 4: The candidate drying laws. W in dB, t in minutes since the start of the tail. The label mn is used when the law generates data (the truth) and Fn when it is fitted to data (the candidate).

Law	Equation	Picture
m1 / F1	$\dfrac{dW}{dt} = -\dfrac{W}{\tau}$	One time constant: exponential drying, the form used in the literature.
m3 / F3	$\dfrac{dW}{dt} = -c\,W^n$	Drying speeds up when the antenna is wetter ($n > 1$); $n = 2$ as truth.
m4 / F4	$W = W_f + W_s,\quad \dfrac{dW_f}{dt} = -\dfrac{W_f}{\tau_f},\quad \dfrac{dW_s}{dt} = -\dfrac{W_s}{\tau_s}$	Two layers of water: a fast one and a slow one that holds a share φ at the start.
m5 / F5	$\dfrac{dW}{dt} = -a\,W - b\,W^2$	A linear and a quadratic term together.
m6 / F6	$\dfrac{dW}{dt} = \begin{cases} -k_{\text{run}}(W - A_c) - k_{\text{evap}}W, & W > A_c \\ -k_{\text{evap}}W, & W \le A_c \end{cases}$	A switch: above a film level A_c water runs off quickly; below it only slow evaporation is left.

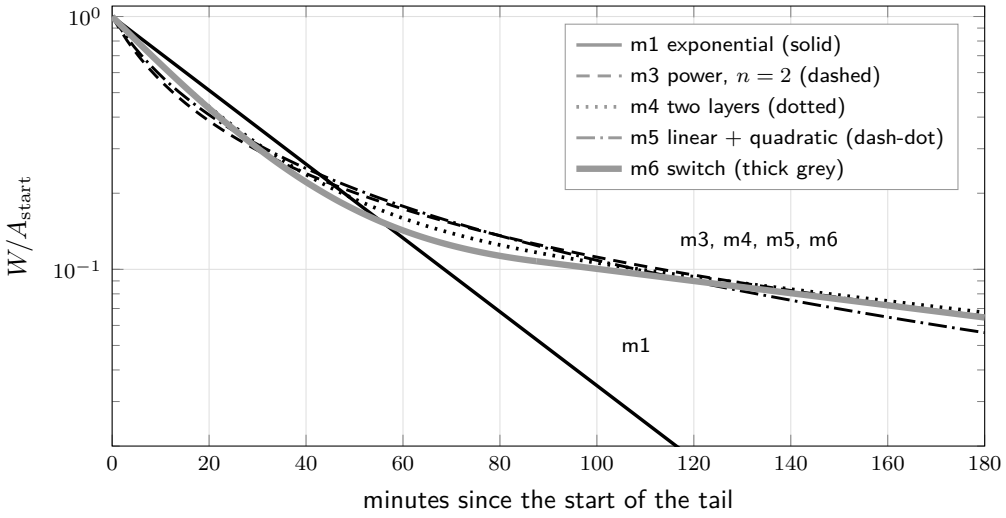


Figure 8: The five laws with the parameters calibrated to the real median tail, for a tail that starts at 1.5 dB, before any tail-to-tail spread or noise is added. On this logarithmic scale the exponential is a straight line; the others bend, and they lie almost on top of each other, which is why they are hard to tell apart. The laws were calibrated together with a spread (Section 6.3), so a single curve is not expected to match the real median by itself.

6.3 The simulator

Tail-to-tail spread

Real tails do not all dry at the same speed: antennas differ, and so do wind, sun and temperature. The simulator gives each tail its own speed by stretching time,

$$W_i(t) = W\left(\frac{t}{f_i}\right), \qquad f_i = e^{\,s\,z_i}, \qquad z_i = \tfrac{1}{\sqrt{2}}\,z_{\text{link}} + \tfrac{1}{\sqrt{2}}\,z_{\text{tail}}, \quad z \sim \mathcal{N}(0,1), \tag{8}$$

so half of the variation belongs to the link (it is the same in all its tails) and half to the single tail. The width s is set so that the band between the 10th and 90th percentile of the simulated tails matches the real band; this gives $s = 1.25$ for every law, i.e. individual tails dry between 5 times faster and 5 times slower than the median.

The measurement chain

A real tail is not $W(t)$; it is $W(t)$ seen through the instrument. The simulator applies the same chain (Figure 9):

$$x(t) = W_i(t) + c_i + n_{\text{slow}}(t) + \sigma_w\,\varepsilon(t), \qquad A(t) = \text{1-minute mean of } \; Q_{0.3\,\text{dB}}\big[x(t)\big], \tag{9}$$

with $c_i \sim \mathcal{N}(0, \sigma^2_{\text{off}})$ a constant baseline error of that tail (from the baseline validation), n_{slow} a slowly wandering noise, ε white noise on the 10-second samples, and Q the rounding of RSL to steps of 0.3 dB.

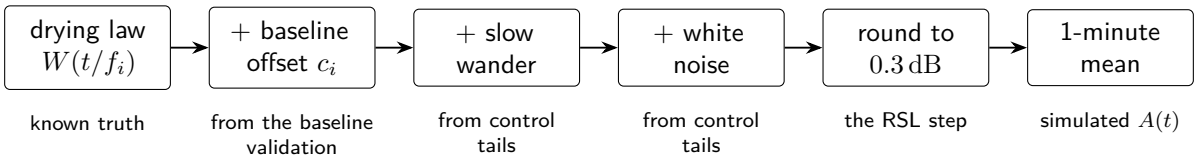


Figure 9: The simulator: a known law seen through the measurement chain. Each block can be switched off to see which distortion hides what.

Getting the noise right

The first version of the simulator had only white noise. Compared with the real control tails (which contain noise and no drying), it was far too quiet over a whole tail: the spread within a tail was 0.05 dB simulated against 0.14 dB real. Real dry-weather noise is correlated in time: it wanders. A simulator without that would have made the laws look much easier to tell apart than they are. The noise model was therefore extended with a slow component (a first-order autoregressive process with its own strength and correlation time), fitted per sub-link to five properties of its control tails. After the change all five agree (Table 5).

Table 5: Real dry-weather noise against the simulated noise (medians over sub-links). The typical fitted noise is a white part of 0.09 dB, a slow part of 0.15 dB and a correlation time of 80 minutes.

Property of a control tail	Real	Simulated
Spread within the tail (dB)	0.141	0.134
Share of minutes exactly on a 0.3 dB level	0.25	0.27
Correlation between minutes 1 minute apart	0.80	0.80
Correlation between minutes 10 minutes apart	0.57	0.63
Correlation between minutes 60 minutes apart	0.22	0.23

Calibrating the laws, and a second lesson

Each law's parameters were set so that its simulated median tail matches the real one (Table 6, Figure 10).

Table 6: Calibrated laws (times in minutes) and how far the simulated median and band are from the real ones (root mean square difference of A/A_{start} over the first 6 hours, after the full measurement chain).

Law	Calibrated parameters	Misfit of the median	Misfit of the band
m1 exponential	$\tau = 29.8$	0.026	0.062
m3 power	$c = 0.053$, $n = 2$	0.044	0.072
m4 two layers	$\tau_f = 18.7$, $\tau_s = 193$, $\varphi = 0.17$	0.033	0.062
m5 linear + quadratic	$a = 0.0030$, $b = 0.045$	0.038	0.069

The important result is in the first row of the table. Before the spread was added, the plain exponential could not produce the slow remainder of the real median, and two layers fitted best; it looked as if the data asked for two layers. With the spread, the noise, the offsets and the real tail lengths included, **a plain exponential reproduces the slow remainder as well as two layers do**. A population of antennas that each dry exponentially but at different speeds looks, in the median, like one antenna with a fast and a slow layer. The shape of the median curve therefore cannot decide between the laws; the decision has to come from the individual tails.

6.4 Fitting one tail at a time: not enough

The natural first method fits every candidate to every tail separately, each with its own start and its own constant offset, and picks the candidate with the best score. The score is the Bayesian information criterion (BIC), which rewards a good fit and charges for every extra parameter. Because the noise wanders, neighbouring minutes are not independent, and a tail of n minutes carries less information than n independent values. This is accounted for by an effective sample size:

$$n_{\mathrm{eff}} = n\,\frac{1-\rho}{1+\rho}, \qquad \mathrm{BIC} = n_{\mathrm{eff}}\,\ln\frac{\mathrm{RSS}}{n} + k\,\ln n_{\mathrm{eff}}, \tag{10}$$

where ρ is the correlation between neighbouring residuals, RSS the sum of squared residuals and k the number of parameters. A typical tail has $n = 210$ minutes but only $n_{\mathrm{eff}} \approx 64$.

Applied to simulated data sets with each of the five laws as the truth (ten data sets per law), the result was negative, and clearly so:

- **Per tail, nothing can be identified.** The exponential wins 60 to 80 % of the tails whatever the true law is. One noisy, quantized tail that starts near 1.5 dB does not contain enough information.
- **Per data set** (scores summed over all tails), only one question can be answered: *single exponential or not.* When the truth is exponential, the exponential wins every time. When the truth is any of the other four laws, the two-layer candidate wins. So if two layers won on real data, it would mean "not a single exponential", and not "two layers".
- The slow time constant of the two-layer law cannot be recovered: the fit trades it against the free offset.
- Switching the distortions off one at a time showed that no single one is to blame. Noise alone, or quantization alone, is enough to hide the small slow layer.

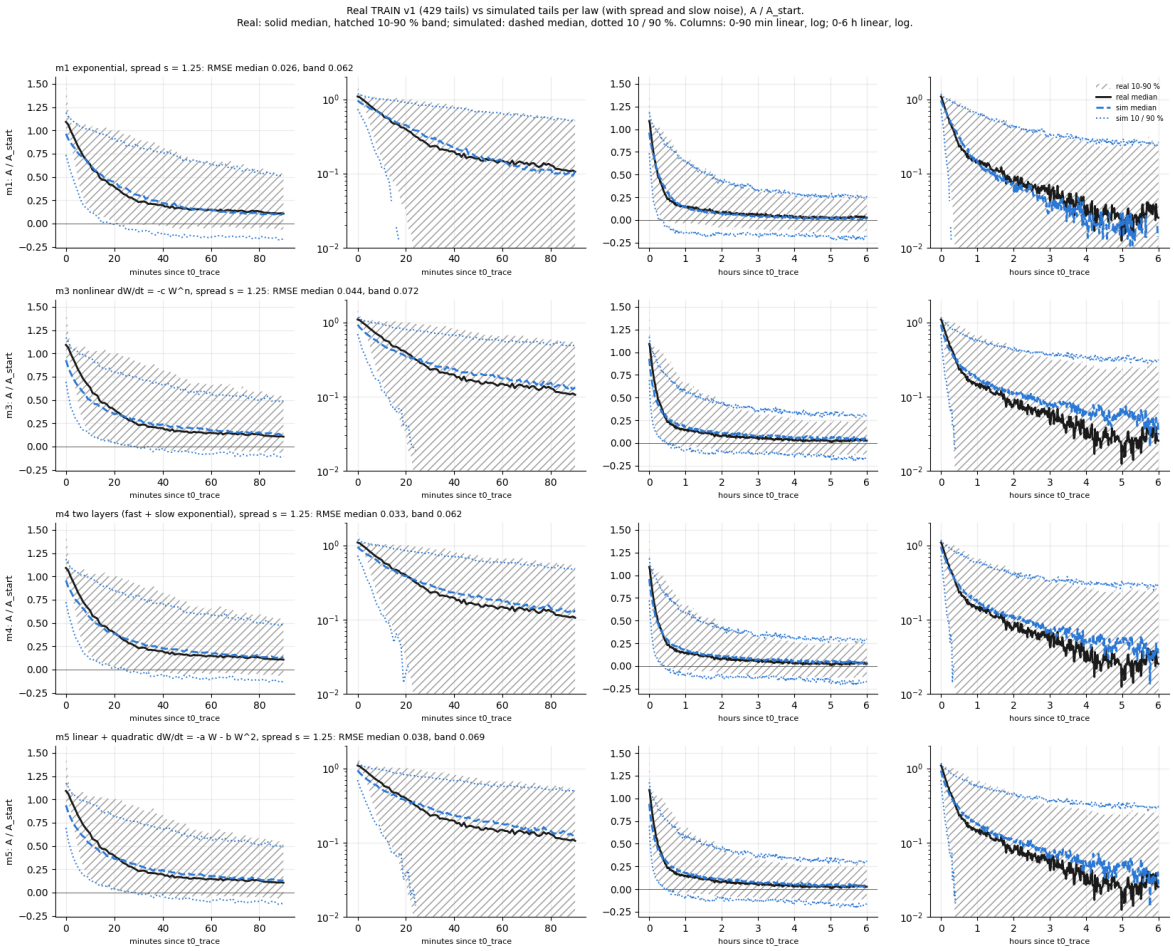


Figure 10: Real tails against simulated tails, one row per law (m1, m3, m4, m5). Solid line and hatched band: real median and 10–90 % band. Dashed and dotted lines: the same for the simulated tails. Columns: first 90 minutes (linear, logarithmic), first 6 hours (linear, logarithmic). All four laws reproduce the real median and band about equally well.

6.5 Fitting all tails together: works on simulated data

The weakness of the per-tail fit is that every tail gets its own free parameters and no tail informs another. But the question is not "what is the law of this tail"; it is "what is the one law that all antennas follow". The second method asks exactly that. One candidate law with one set of parameters θ is fitted to the whole data set at once:

$$A_i(t) \;=\; W\!\left(\frac{t}{f_i};\ \theta,\ W_{0,i}\right) + c_i + e_i(t). \tag{11}$$

The pieces, and what each one fixes compared with the per-tail fit:

- θ: the law's parameters, **shared by all tails**. This is where the information is pooled.
- $W_{0,i}$: the start of tail i, free.
- $f_i = e^{\,s(u_{\text{link}}+v_{\text{tail}})}$: the speed of tail i, as in Eq. (8). It is not fitted per tail. It is treated as a random quantity with a width s that is fitted, and the likelihood is *averaged* over all the speeds a tail could have. This is what separates "many exponentials with different speeds" from "two layers".
- c_i: the offset of tail i, no longer free. It has the prior $c_i \sim \mathcal{N}(0, \sigma_{\text{off}}^2)$, with σ_{off} the baseline error of that sub-link measured in dry weather. The offset can no longer absorb a slow layer for free.
- $e_i(t)$: the noise, with the white and the slow part of that sub-link measured on its control tails. The likelihood uses this correlated-noise model directly, instead of correcting for it afterwards.

Schematically, the quantity that is maximised over θ and s is

$$\mathcal{L}(\theta, s) \;=\; \prod_{\text{links}} \int du\ p(u) \prod_{i\,\in\,\text{link}} \int dv\ p(v)\ \max_{W_{0,i},\,c_i}\ p(A_i \,|\, \theta, W_{0,i}, c_i, f_i)\, p(c_i), \tag{12}$$

and the candidates are compared by $\text{BIC} = -2\ln\mathcal{L} + k\ln N$, where N counts the independent units (a link in an event), not the minutes and not the tails.

The test and its result

The test data follow the structure of the real set. Ten data sets per true law were built by an *event bootstrap*: 24 events are drawn with replacement from the 24 real ones, with all their tails, so that the dominance of a few events is part of the test. A rule for success was written down before the run: the method is useful if the true law's own candidate wins in at least 80 % of the data sets for the exponential, two-layer, linear + quadratic and switch laws, and the main parameters are recovered within ±30 %.

The rule was met, using the first 60 minutes of each tail, the first 6 hours, or the whole tail (Figure 11):

- The right law wins in 80 to 100 % of the data sets for the four laws in the rule.
- The parameters come back within about ±15 % (median), including the slow time constant and the share of the slow layer, which the per-tail fit could not recover.
- One confusion remains: the pure power law (m3) is taken for the linear + quadratic law (F5) in 20 to 40 % of its data sets. The two are close relatives, and F5 contains a quadratic term.

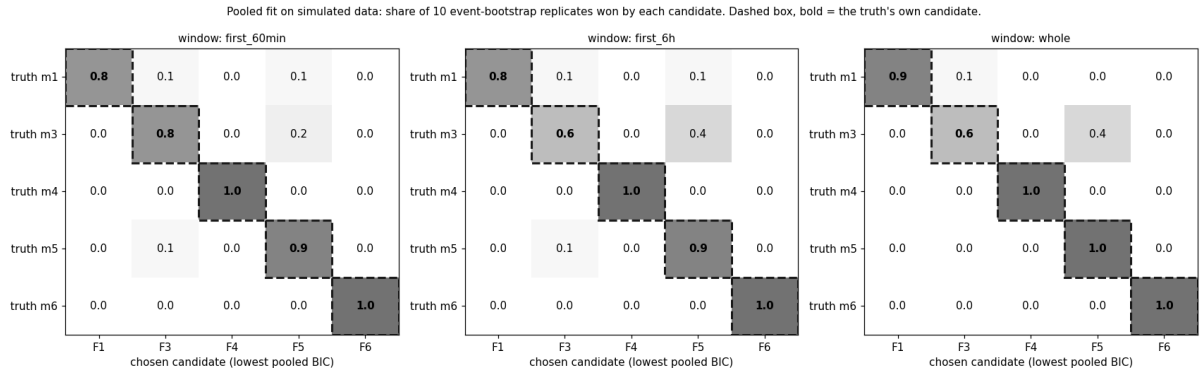


Figure 11: Pooled fit on simulated data. Rows: the law that generated the data. Columns: the candidate chosen. Each cell is the share of the ten data sets; a perfect method would have 1.0 on the diagonal (dashed boxes). Panels: first 60 minutes, first 6 hours, whole tail.

What this does and does not show

On the real tails, the pooled fit will be able to say which of the five forms describes the drying best, with its parameters and an uncertainty from resampling events. It will not prove a mechanism: the true law may be none of the five, and the winner is then only the best approximation among them. And the test above was run under friendly conditions, which is why the next step was added:

- the fit used the same noise model that generated the data, while the real noise has more than one time scale;
- the search started from the calibrated parameters of each law;
- the real tails decay faster in the first hour than every calibrated law (real median A/A_{start} at 30 minutes is 0.23; the simulated laws give 0.28 or more). This is a hint, not a result: the uncertainty of the real value is wide (0.16 to 0.56).

6.6 Stress test of the pooled fit: partly passed

The stress test tries to break the pooled fit by removing the three friendly conditions one at a time, again on simulated data and again with a rule for success fixed before the run (per test and per true law: the law's own candidate wins in at least 4 of 5 data sets, and its main parameters come back within $\pm 30\,\%$). It uses the first 60 minutes of each tail and four true laws: exponential, two layers, linear + quadratic, and the switch.

T1, wrong noise model.
The data are generated with two slow noise components (correlation times of 20 and 300 minutes), but the fit still assumes one. Slow noise that the fit does not model could be mistaken for a slow layer and favour the more flexible laws.

T2, honest starting points.
The search starts from four generic values, the same for every true law, instead of from the right answer; the best of the four is kept.

T3, faster truths.
Every law is recalibrated to the real median at 10, 30 and 60 minutes, so that the test data dry as fast as the real tails do. The exponential and the two-layer law are also tested with a wider spread of drying speeds ($s = 1.5$).

Table 7: Stress test of the pooled fit on simulated data: 12 of the 14 cases pass. "Pass" means the rule fixed before the run is met.

True law	T1: wrong noise	T2: honest starts	T3: faster truths
m1 exponential	pass	pass	pass (also with the wider spread)
m4 two layers	pass	pass	pass (also with the wider spread)
m6 switch	pass	pass	pass
m5 linear + quadratic	**fail**: the power law wins 5 of 5	pass	**fail**: wins 4 of 5, but its linear rate comes back at 0.65 of the truth

What the result means:

- **The method holds for three of the four laws.** The exponential, the two-layer law and the switch are recognised under noise the fit does not know, from generic starting points, and when they dry as fast as the real tails. The fear that unmodelled slow noise would be mistaken for a slow layer did not come true in this test.
- **The two nonlinear laws are one family.** Both failures belong to the linear + quadratic law, and both have the same cause: with the calibrated parameters its linear term is small, so it is almost a pure quadratic, which the power law copies. The margins are tiny (about 2 BIC units, in either direction). On real tails the two will be read together as "drying that speeds up when the antenna is wetter", and the linear rate will not be interpreted from the first hour.
- **Starting points matter, most of all for the switch law.** From different starting points the switch candidate reached clearly different solutions in all 20 data sets of the test, the other candidates in about a fifth to a half of them. The winner still was the same as with the ideal start in 19 of 20 data sets. Every candidate will therefore be fitted from several starting points on the real tails.
- **Two layers against the switch needs a clear margin.** In one two-layer data set the switch won from a generic start by about 4 BIC units; a difference below about 10 units between these two will not be called a decision.

One limit of the test itself: when the slow laws were recalibrated to the fast early decay with all parameters free, their slow part ran off to a constant floor, so the slow parameters were kept at their earlier values. This choice was made before any fit of T3 and is still to be confirmed by us.

6.7 SINDy directly on the tails: failed

The second tool of the proposal is weak-form SINDy [10]. It writes dA/dt as a sum of terms from a library (here $1, A, A^2, A^3$ and switch terms $\max(A - A_c, 0)$), keeps only the terms it needs, and avoids differentiating the noisy signal by integrating it against smooth test functions over windows of 21 minutes. It was run on the simulated data sets of the stress test, with the rule for success fixed before the run. It failed all four questions asked of it.

- **It returns "A" for almost every truth.** In the first hour, with the spread of drying speeds, the quadratic and the linear + quadratic truths give the linear law in 5 of 5 data sets, with no A^2 term. The switch truth gives the linear law with no switch term. The exponential truth keeps its term but gets a false A^2 term in 3 of 5 data sets and a rate 1.8 times too high.

- **A hidden second state is not noticed.** The two-layer truth returns a clean single-variable law in 5 of 5 data sets, which would be read as a finding.
- **It disagrees with the pooled fit.** The two tools name the same family of law in 1 of 20 data sets; the pooled fit is right in 19 of 20.

The reason is the two things the pooled fit models and SINDy on raw tails does not: the baseline offset of each tail and the spread of drying speeds. With the true offset given and no spread, SINDy does find A^2 in 5 of 5. The spread alone creates the *mixing trap*: exponential tails of different speeds, put into one regression, look like a nonlinear law (a false A^2 in 3 of 5 data sets with the spread, in none without it). So SINDy will not be run on the raw real tails as first planned.

6.8 Two jumps: draw the drying curve, then name it

Discovery was kept but split in two (Figure 12).

Jump 1, draw the curve.
The pooled fit of Eq. (11) is used again, but the menu of laws is replaced by a free smooth curve $k(W)$, the drying rate per dB of water:

$$\frac{dW}{dt} \;=\; -f_i\, W\, k(W), \tag{13}$$

with f_i the speed factor of tail i. The curve is a spline with five knots, and how smooth it must be is chosen by leaving out half of the events. A flat k is exponential drying; k proportional to W is quadratic drying. On a log-log plot the slope of k is therefore 0 for the exponential law and 1 for the quadratic one.

Jump 2, name it.
Sparse regression is applied to the clean curve $h(W) = W\,k(W)$ with the library W, W^2, W^3 and switch terms. Offsets, spread and noise were dealt with in jump 1, so the mixing trap should not appear.

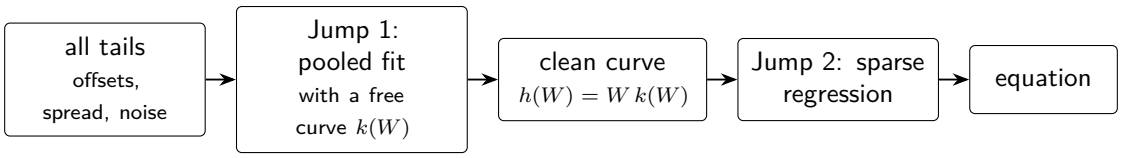


Figure 12: The two jumps that replace SINDy on raw tails.

First run. By the rule fixed before the run the result was only partial. The curve is badly determined at low W: for the exponential truth its lowest point came out 2 to 4 times too high in 9 of 10 fits. Looking at the results afterwards, the curve and its equation were right above about 0.35 dB. A finding made after seeing the results is not a result, so it was tested again.

Confirmation run. New simulated data, the first 60 minutes, and a judged range fixed in advance: from 0.35 dB up to the highest level with enough data (2 to 3.5 dB). Nothing below 0.35 dB is judged or named. Table 8 gives the result.

Table 8: Confirmation run of the two jumps on fresh simulated data (5 data sets per law, first 60 minutes, W above 0.35 dB). "Slope" is the slope of $\log k$ against $\log W$.

True law	Slope: true	Slope: found	Equation found (coefficient / truth)
m1 exponential	0	−0.01 to 0.05	W (×0.99), one term in 5 of 5: pass
m3 quadratic	1	0.99 to 1.12	W^2 (×0.97), one term in 5 of 5: pass
m5 linear + quadratic	0.92	0.93 to 1.03	W^2 (×1.13), one term in 5 of 5: pass
m6 switch at 0.16 dB	0.23	0.20 to 0.32	three terms, not a law (not judged: the switch is below the range)
m4 two layers	–	0.36 to 0.47	three terms; the warning sign fired in 2 of 5: fail

For the three single-curve laws the true curve lies inside the band of the five fits over the whole judged range, and the mixing trap does not appear. Two limits remain. A switch below the range and a two-layer law both give a *mildly rising* curve (slope 0.2 to 0.5) with a many-term equation, and the curve cannot say which of the two it is. And the hoped-for warning sign for two layers (the free curve fits clearly worse than the two-layer candidate) fired in only 2 of 5 data sets, so it is not used as a check.

6.9 The rest of the method, and a false step

Four additions were tested on the same simulated data, each with its own rule fixed before the run (Table 9).

Table 9: Tests of the remaining parts of the discovery method, on simulated data.

Part	What happened	Used on real tails?
Naming alone: the true curve plus 2 to 5 % noise	Sparse regression repeated on resampled points names exactly the true terms in 20 of 20 draws for all four laws, the switch included. Symbolic regression (PySR [7]) does so in 20, 20 and 18 of 20 for the exponential, quadratic and linear + quadratic laws; it cannot write a switch. Choosing the best subset of terms by BIC adds one extra term in 3 to 8 of 20 draws.	yes
A curve that may contain a step (to find a switch)	A step is kept in 5 of 5 data sets for *every* truth, also the pure exponential.	no
Choosing between candidate laws by the likelihood of left-out events	The law with the false step wins in 18 of 20 data sets. Comparing only the five named laws by BIC on the whole data set still names the right family in 4 or 5 of 5.	no; BIC of the named laws instead
An uncertainty band from resampling the events of one data set	The band is ±1 to 6 % wide and contains the true curve on 0 to 78 % of the range (rule: 80 %). It is about three times too narrow: the level of a fit is off by −7 to +23 %. The slope is steadier, within 0.01 to 0.11 of the truth.	shown, with this warning

Is the quantization the cause of the false step? The first explanation was that RSL is recorded in steps of 0.3 dB, which the simulator copies but the noise model of the fit does not describe. This was tested directly: exponential data were simulated three times with the same random numbers, with no rounding, with 0.3 dB and with 0.1 dB. The step is kept in 5 of 5 data sets in all three cases, also with no rounding at all. So the quantization is *not* the cause, and adding it to the fit is not expected to help. The test showed one more thing: in 15 of 15 fits the step stays where the search started it, so its position carries no information either. The cause is

open.

What can be said on real tails, as a result. For W between 0.35 dB and the top of the data, in the first hour after rain: how the drying rate per dB of water changes with W, as a slope known to about ± 0.1; whether that stretch is a one-term law (exponential or quadratic); the level of the rate to about $\pm 20\,\%$; and which of the named laws the data prefer by BIC. Nothing can be said below 0.35 dB, and nothing about a switch level: a step found there would mean nothing.

In short. A simulator that copies the real tails and their noise shows that the median curve cannot decide between the laws, and that fitting single tails cannot either. Fitting all tails together, with a shared law, a random speed per tail, a bounded offset and a correlated-noise model, recovers the right law and its parameters on simulated data. SINDy on the raw tails does not. Drawing a free drying curve with the pooled fit and then naming its equation does, for W above 0.35 dB in the first hour. A step in the curve, a choice by held-out likelihood and a resampling band failed their tests. All of this is on simulated data; the first fit of the real tails is the next section.

7 The first fit of the real tails

7.1 What was done

This is the first step that fits a law to real data. It uses the frozen tail set, training events only: 429 tails from 24 events on 159 links. The 123 validation tails were not read. Only the tools that passed on simulated data were used, with their settings unchanged, and the rule for reading the result was written down before the run:

- the five named laws, each fitted to all tails together from four starting points and compared by BIC;
- the free drying curve $k(W)$ of Eq. (13), drawn from the first 60 minutes of the tails and read between 0.35 dB and 2.83 dB (the top of the data), with a band from 20 refits on resampled events;
- the naming of that curve by sparse regression, with symbolic regression as a cross-check.

Every result was repeated on other versions of the set: without the events flagged for many rising tails, without the tails with a level flag, on the alternative set with weaker tails, and with each of the three largest events left out in turn.

7.2 What was found

No drying law is identified. The free curve *falls* as W grows: the slope of $\log k$ against $\log W$ is -0.87 on the main set (-1.43 to -0.82 over the resampled refits). In words, a wetter antenna would dry *more slowly* per dB of water: the time constant $1/k$ is 6 minutes at 0.35 dB, 33 minutes at 1 dB and 71 minutes at 2.83 dB. None of the simulated truths has this shape: an exponential gives slope 0, and the nonlinear laws give a rising curve. A direct look at the measured rate of decrease, with no model, shows the same trend, so it is in the data and not made by the fit.

Where it comes from. Many tails do not decay at first. Ten minutes after the tail starts, 21 % of the tails are still at 90 % or more of their starting level. In the events that had been flagged for many rising tails this share is 38 %; in the other events it is 11 %. A tail that stays flat at a high level tells the fit that strong attenuation dries slowly. Table 10 shows how much the result depends on which tails are included.

Table 10: Slope of the free drying curve on the real tails, for different versions of the tail set. Slope 0 means an exponential law.

Tails used	**Slope of** $\log k$ **against** $\log W$
Main set (429 tails)	-0.87
Without the events flagged for many rising tails	-0.05
Without the tails with a level flag	-0.79
Alternative set (weaker tails included)	-0.69
One of the three largest events left out, in turn	-0.60, -1.60, -0.46

Without the flagged events the curve is flat, which is the exponential law, with a time constant of about 15 minutes. So the answer depends on a choice that has not been made: which tails count as drying.

The curve is badly determined. The four starting points of the fit end far apart (800 to 1700 units of the fit's cost; on simulated data the difference was about 8), and the resampling band is $\pm 94\,\%$ wide. A dip near 0.9 dB moves when a knot of the curve is moved, so it is not read.

No equation. Sparse regression keeps all five terms of its library, which means it finds no short formula. The best formula with one term misses the curve by 61 % (root mean square), and the simplest formula from symbolic regression is a constant rate, which misses it by 49 %.

The named laws, and a fault in the comparison. As run, the exponential law wins by BIC, with a time constant of 25.2 minutes. Looking into this found a fault in our own tool: for the exponential and the two-layer law the starting level of a tail may be fitted as negative, while the other laws and the free curve keep it positive. Three rising tails with a negative start account for the whole advantage of the exponential over the power law with exponent 1, which is the same law. Compared like for like, the switch law is ahead of the exponential by 21.5 BIC on the main set; over the other versions of the set this ranges from the exponential ahead by 0.9 to the switch ahead by 216. This is not decisive either. The residuals of the fit have 4.1 times the variance that the dry-weather noise model allows, so every BIC difference is overstated, and the switch law depends on its starting point. One thing is stable in all seven versions of the set: the power law ends at exponent 1 and the linear + quadratic law ends with no quadratic term. A drying rate that *rises* with W is therefore not supported in the first hour.

Shape of the real tails. The median of A/A_{start} is 0.568, 0.229 and 0.142 at 10, 30 and 60 minutes.

Literature. The comparison with a published drying time constant could not be made: neither the proposal nor the other documents of the project quote one, and none was invented. The values have to be read from [1] and [9].

7.3 How we read it

The model that was tested, $dW/dt = -h(W)$ with one curve for all tails and a random speed per tail, does not describe the first hour of the real tails. A tail that stays flat or rises after the last trace of rain points to something the model leaves out: water still arriving (light rain or drizzle below what the references see, or condensation), which belongs to the wetting term $g(R, W)$ of Eq. (19), or a tail that starts too early. The exponential with a time constant of 15 to 28 minutes is a placeholder, not a discovered law.

Three decisions are open before the next step: (a) which tails define drying (all of the frozen set; the set without the flagged events; or tails restarted where the decay begins, which would be a new frozen set); (b) repairing the fault in the starting level and repeating the tests on simulated

data, this time with some rising tails among them; (c) a larger noise level, or a fitted one, for drying tails.

In short. The first fit of the real tails did not find a drying law. The drying rate falls with W (slope -0.87), because about one tail in five stays flat or rises after the rain. Without the events where this is common the tails are close to an exponential with a time constant of about 15 minutes. No short equation fits the curve, the comparison of the named laws is not decisive, and a fault in that comparison was found. A drying rate that rises with W is not supported.

8 Stage 1, part 3: what is in a drying tail, and why the tails were left (done, closed)

The first fit of the real tails left three decisions open (which tails define drying, a repair of the tools, the noise level). Before taking them, four steps asked what a tail actually contains. None of them fits a drying law; each rule was written down before its run, on training events only.

8.1 Why some tails wait or rise (T15)

The 429 tails were sorted into four classes by their shape in the first hour: 240 decay at once, 64 stay on a plateau and then decay, 36 rise and 89 stay flat or fall slowly. The share that does not decay at once varies from 0 to 100 % between events. Two explanations were tested against each other: something still wets the antenna after the last trace of rain, or a law of the antenna itself whose rate per dB falls as W grows. Neither passed its rule. What the data do show: waiting is shared by the other links of the same event at the same time (1.85 times more than by chance), the two directions of a link agree in 81 % of the tails (64 % by chance), it does not depend on the path length, and half of the rising tails are at night (8 % of the decaying ones). After its plateau, a waiting tail follows the same curve as the decaying tails. So the falling drying curve of Section 7 comes from mixing tails that wait with tails that dry; waiting is a property of the *event* (the weather at that time), not of the level, the link or the length. What causes it stayed open.

8.2 What is in a tail (T16)

Three questions were asked without any law.

- **Is a rise water at all?** Each rising tail was compared with the same sub-link at the same clock time on certified dry days. 24 of the 36 rising tails rise more than 95 % of their own dry days: most rises are water, not a drift of the baseline with the daily cycle, so no new baseline was needed. Whether the water is dew or drizzle could not be decided, because the humidity record is missing for the two events with most rising tails.
- **When does drying start?** The rain references within 5 km of the link fall silent about 25 minutes after the tail starts (median), and by then 349 of the 429 tails are already below 1 dB. Most of what was called a drying tail happens while it still rains nearby.
- **Antenna or path?** The attenuation per minute was regressed on the path length around the start of the tail. This test had no power: the rain at the end of an event is so light that the radar expects only 0.03 to 0.06 dB per km, inside the uncertainty.

8.3 Does drying start when the nearby rain ends (T17)?

From the moment no reference within 5 km shows rain, the attenuation halves in about 40 minutes. A placebo start that is just as late but ignores the rain gives 37 minutes (difference +3 minutes, interval −2 to +7). By the rule this is the *same* speed: the end of the nearby rain is not a visible event in the signal. A tail has two phases in time, a fast one in the first 15 to 20 minutes and a slower remainder that halves in about 40 minutes.

8.4 Very short links (T18)

Rain on the path grows with L; water on the antennas does not. The fast first drop of a tail was therefore compared between short links (below 1 km) and long ones (2 km and more), matched on event and starting level (Figure 13). The relative drop is 0.41 on short and 0.44 on long links, a ratio of 0.94 (interval 0.84 to 1.15). Path rain alone would give a ratio of about 0.2. By its pre-set rule the result is *undecided*, because the radar expects almost no path rain before a tail, so the test cannot reach the precision the rule asked for. Read beyond the rule, the fast drop does not scale with the length over a factor of ten, which speaks for the antenna (or for anything else that does not scale with L). The step also showed that there are enough short links for the next phase: 41 training links below 1 km with 1663 (event, sub-link) pairs, all at 28 and 38 GHz, and that the peak attenuation on them follows the peak radar rain closely (rank correlation 0.85).

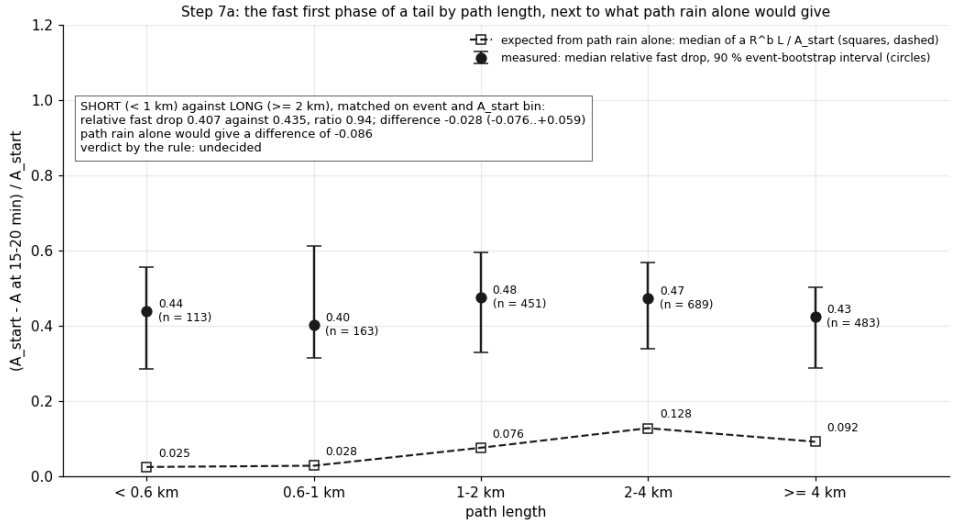


Figure 13: The fast first drop of a tail, relative to its starting level, by path length (circles, solid bars: measured, with 90 % intervals). Squares, dashed line: what rain on the path alone would give. The drop does not grow with L.

8.5 The decision to stop studying tails

On 13 October 2026 the tail approach was closed, and the planned steps that built on it (what drying depends on, a wetting law, the verification of a drying law) were cancelled. The reasons, taken from the steps above:

- a tail starts after the *last* shower of an event that lasts about 9 hours and has several showers; by then A is already at about a fifth of its peak;
- there is only one tail per event, and its start is uncertain;
- wetting and drying were split artificially: most of the fall of A happens while the rain fades;

- there was no criterion of success, so the checks did not converge.

What was kept: on short links A is mostly WAA in light and moderate rain, WAA follows the rain within minutes, there are two phases after rain, rises are water, and there are enough short links to work on whole events.

In short. Tails wait or rise because of the event, not the link; most of a tail passes while it still rains nearby; the fast first drop does not grow with the path length. The drying tail was the wrong object, and the tail approach was closed on 13 October 2026.

9 Stage 1, part 4: system identification on whole events (S1–S4.3)

9.1 The idea

Instead of cutting tails, each whole event on a link is treated as a system: the radar rain along the path, $R(t)$, is the input and the attenuation $A(t)$ is the output (Figure 14). The model is

$$A_{\mathrm{pred}}(t) \;=\; W(t) \;+\; L\,P_{\mathrm{band}}\big(R(t-d)\big) \;+\; c, \tag{14}$$

where $W(t)$ is the WAA, the state whose law is sought; $L\,P(R)$ is the path part, with P a free curve per frequency band that starts at $P(0) = 0$ and never falls; d is a small time lag of the radar; and c is an offset per event and sub-link, taken from the dry minutes before the rain and never fitted. The antenna part is driven by $a_{\mathrm{link}}\,F_{\mathrm{band}}(R)$: a free level curve F per band times one amplitude per link.

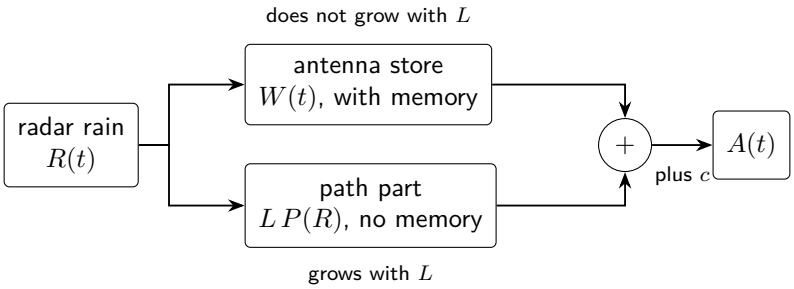


Figure 14: The system view. The two parts are told apart only by how they depend on the path length L.

The standing rule. WAA is never estimated by subtracting a power-law rain term. If it were, every error of the assumed power law would end up in the WAA. The path part is separated from the antenna part only through its dependence on L at matched rain: the antenna part is the same on a 0.3 km and on a 5 km link, the path part is not. The standard power-law coefficients are used only to build fake data and as a size reference on figures.

The model ladder. Each rung adds one thing to the one before:

- M0: no WAA, $W = 0$.
- M1: static, $W = a_{\mathrm{link}}F(R)$.
- M2: one memory, $\dfrac{dW}{dt} = \dfrac{a_{\mathrm{link}}F(R) - W}{\tau}$.
- M3: separate speeds, $\tau = \tau_{\mathrm{wet}}$ while W rises and τ_{dry} while it falls (a dynamic wetting and drying model in the spirit of Schleiss et al. [1]).

- M4: two stores, $W = W_f + W_s$. The fast store goes towards $\varphi\, a_{\text{link}} F(R)$ with its own wetting and drying times; the slow store goes towards $(1-\varphi)\, a_{\text{link}} F(R)$ with one time constant τ_s. M4b is the same with τ_s bounded to 30–240 minutes.
- M5: a free law $dW/dt = g(R, W) - h(W)$, to be named afterwards by sparse regression (SINDy [6]) and symbolic regression (PySR [7]).

The judge and the stop rule. Each model is judged by its prediction error of A on events it was not fitted to (leave one event out), with intervals from resampling whole events. The answer is the *simplest* model whose difference to the best has an interval that contains zero. Before any rung is read on real data, a *fake check* must pass: data made from a known truth, driven by the real rain, lengths and noise, must give that truth back.

9.2 S1: the ladder on links below 1 km

On 41 links below 1 km (1718 events on a sub-link) the fake check passed: the true rung was chosen every time and the time constants came back within 10 %. On the real data every rung with memory beats the static one, and the two-store model M4 is the best, but the whole ladder removes only 6 % of the error of "no WAA" (prediction error 0.417 dB for M0, 0.390 dB for M4). The stable number is the fast drying time, about 8 minutes in every run. The wetting time is *not* determined: 17 minutes with the radar as input, 2.4 minutes with a gauge near the link. The slow store was not identified (its time constant ran to the bound), and the split between path and antenna was not firm from links this short.

9.3 S2 and S2b: links up to 3 km, the free rung, and two repairs; WAA v2

S2 extended the data to links up to 3 km (6653 events on a sub-link) and added the free rung M5. Four fake checks came first, and two of them failed:

- **Input smoothing (passed).** With the exact 1-minute rain the fit recovers a wetting time of 3 minutes; with a 5-minute, noisy rain input like the radar it reports 6.3 minutes, 2.1 times slower. The slow wetting seen with the radar is the input, not the antenna.
- **Drift or slow store (passed).** A slow level drift and a slow water store can be told apart.
- **Path and antenna curves (failed).** The path curve came out near zero at light rain even with the true rain as input. The cause was found: an optimiser trap in the way the curves were parametrised.
- **The free rung (failed).** M5 predicted as well as M3 but returned levels about half the truth: its surfaces are not identified, and no formula could be named from the real data (the named law let W run away).

S2b repaired the trap (the curve check then passed: all knots within 15 % of the truth), bounded the slow store (its check passed too) and gave the fits more iterations. On the real data the trap had hardly mattered. By the stop rule the one-store model M3 was chosen on the short links: **WAA v2**. With a gauge as input its wetting time is 2.9 minutes (interval 2.5 to 3.6) and its drying time 12.8 minutes (11.3 to 15.5). A slow remainder after the rain, 2 to 4 % of the peak over 1 to 2 hours, is described by no rung.

9.4 S3 and S3b: verification, and a model-free length test

The model-free length test. In steady rain the WAA has reached its level, so for every frequency band and rain class the attenuation of all links can be fitted against their length,

$$A - c \;=\; I \;+\; s\,L, \tag{15}$$

over links of every length (training events, 5-minute bins with steady rain, resampled by events). The intercept I is the part that does not grow with L, the antenna part; the slope s is the path part per km. This needs no model at all. Table 11 and Figure 15 give the result.

Table 11: The model-free intercept I and slope s of Eq. (15), training events, with 90 % intervals, and $L_0 = I/s$, the extra length that a constant WAA would look like (Eq. (4)).

Band	Rain (mm/h)	I (dB)	s (dB/km)	L_0 (km)
28 GHz	0.5–1	1.02 [0.84, 1.26]	0.08 [0.05, 0.12]	12.0
28 GHz	1–2	1.33 [1.07, 1.58]	0.18 [0.14, 0.22]	7.5
28 GHz	2–5	1.86 [1.52, 2.12]	0.34 [0.28, 0.42]	5.5
28 GHz	5–10	2.45 [2.08, 2.78]	0.82 [0.72, 0.93]	3.0
28 GHz	10–20	2.82 [1.64, 5.38]	1.44 [0.68, 2.13]	2.0
38 GHz	0.5–1	1.10 [0.91, 1.36]	0.16 [0.11, 0.20]	7.0
38 GHz	1–2	1.47 [1.13, 1.84]	0.36 [0.29, 0.42]	4.1
38 GHz	2–5	2.08 [1.74, 2.39]	0.64 [0.47, 0.78]	3.3
38 GHz	5–10	2.35 [2.00, 2.81]	1.56 [1.20, 1.70]	1.5
38 GHz	10–20	2.44 [2.02, 3.99]	2.66 [2.24, 3.09]	0.9

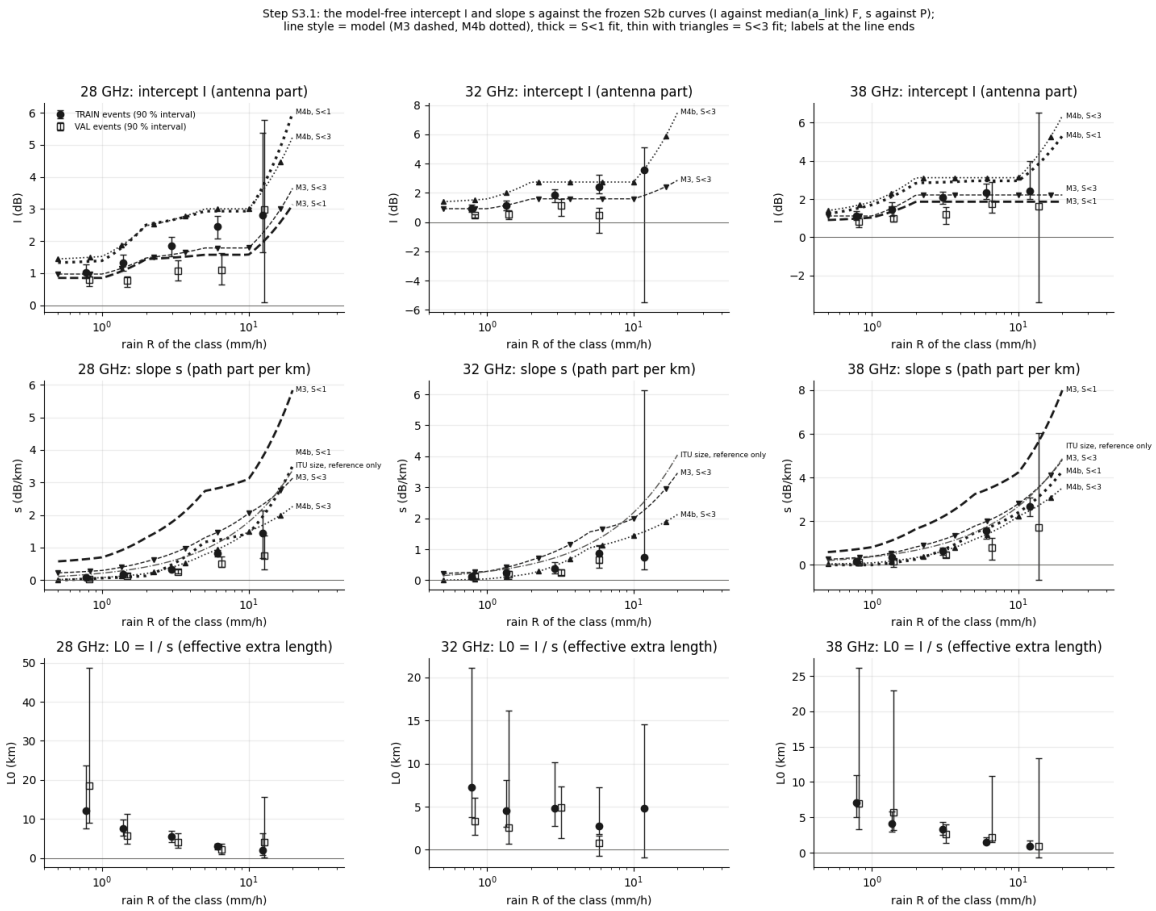


Figure 15: The model-free intercept I (top), slope s (middle) and $L_0 = I/s$ (bottom) against the rain rate, per band. Filled circles: training events; hollow squares: validation events; bars: 90 % intervals. The lines are the models of S2b (dashed M3, dotted M4b, labelled at their ends) and the standard power law as a size reference only.

What it says: the antenna part rises with the rain and then saturates at about 2 to 3 dB, while the path part keeps growing. L_0 therefore falls from about 12 km at light rain to about 1 to 2 km in heavy rain: the WAA is *not* a constant extra length, and $f = L + L_0$ with one L_0 does not describe the data. This is the first direct test of the trap of Eq. (4). Neither WAA v2 (M3) nor M4b passed this test (in 0 and 2 of 10 cells): M3 hides antenna wetting in its path curve, and M4b's path curve is too small at light rain.

Unseen events. On the 18 validation events, read here for the first time with frozen models, the overall prediction error is no worse than on training (M4b 0.360 dB on links below 1 km). The two directions of a link agree on their peaks (rank correlation 0.96). With a gauge as input, refitted on the validation gauge links, the wetting time is 1.8 minutes and the drying time 10.8 minutes: the timing of WAA v2 is replicated.

Fitting on all lengths (S3b). Because the path curve cannot be pinned by links below 1 km, the same models were fitted on the training events of *all* link lengths (up to 8.7 km), once free and once with the wetting and drying times fixed at the gauge values. With all lengths the path curve is pinned, and the fixed-timing models reproduce the model-free I and s on training in 10 of 10 cells. On the validation events, however, every model failed (0 to 1 of 8 cells): there the observed I and s are only 0.72 and 0.61 times what the models predict. The deficit is common to the antenna and the path part, which points to a difference between the two sets of events, not to the split between antenna and path. The residual during rain is flat with L on validation

(the scaling with length is right) but about -0.1 to -0.2 dB at every length.

9.5 S4.1 to S4.3: why the validation events differ

Is it ordinary spread between events? The split of events into training and validation is random within each intensity class. Of 200 random halves of the training events, only 0 to 0.5 % fall as low as the validation set. The gap is more than ordinary spread.

Per-event factors (failed check). One rain scale k_e and one antenna factor α_e per event were fitted on the frozen model. Their fake check failed: only 63 % of the events recovered both (rule 80 %). Heavy events passed (16 of 16); with little rain, k_e and α_e trade against each other. The factors were therefore not interpreted, and none of 28 tests against the weather (temperature, wind, humidity, dryness before the event) was a discovery.

Radar against gauges. Without any factor, the radar was compared with the City gauges on the events with enough gauge rain: the radar reads 1.71 times the gauges on the validation events (5 events) and 0.89 times on the training events (14 events). Their ratio, about 0.5, is the size of the gap.

S4.3: the test of that lead. Per event, the radar rain was divided by its own radar-to-gauge ratio g_e, taken from the gauges only and never from the links, and the length test of S3b was repeated with the frozen model and no refit. The rule was written before the run: the gap is "explained" if the median observed-to-model ratios of I and s lie within 0.85–1.15 and both lie inside their intervals in at least two thirds of the cells, and the training set must still pass the same rule. Table 12 and Figure 16 give the result.

Table 12: The length test through the frozen model (fast and slow store, gauge timing) before and after dividing the radar rain by the event's radar-to-gauge ratio. Ratio = observed / model, median over the eligible cells.

Events	Rain input	Ratio of I	Ratio of s	Both inside
Validation	radar as it is	0.72	0.61	1 of 8
Validation	radar / g_e (path mean)	0.99	0.95	5 of 6
Validation	radar / g_e (gauge pixel)	1.00	0.97	5 of 6
Validation	radar / one ratio per set	0.94	0.99	6 of 6
Training	radar as it is	1.07	0.99	10 of 10
Training	radar / g_e (path mean)	1.11	0.92	7 of 10

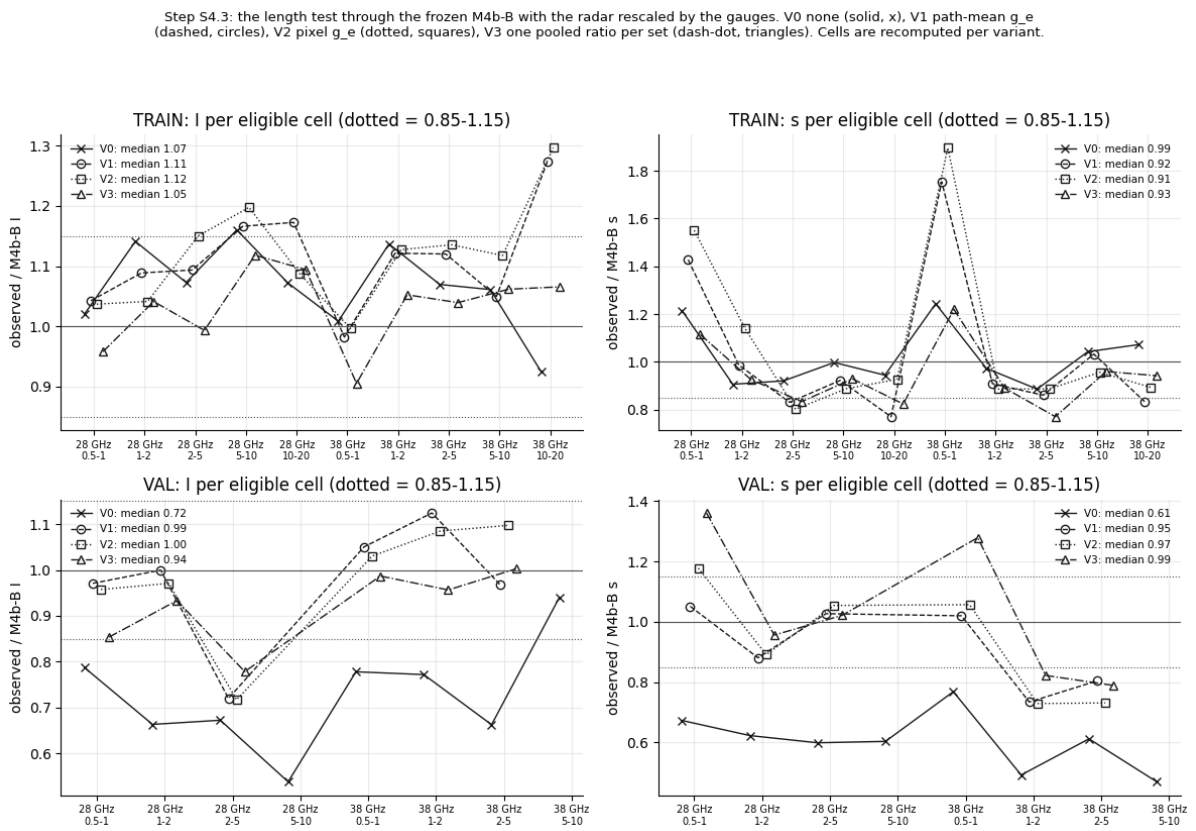


Figure 16: Observed / model ratio of I (left) and s (right) per cell, training (top) and validation (bottom). Line style and marker give the rain input: solid with crosses as it is; dashed with circles divided by the event's ratio (path mean); dotted with squares the same from the gauge pixels; dash-dotted with triangles one ratio per set. The dotted horizontal lines mark 0.85 and 1.15.

By the rule the validation gap is **explained by the rain reference**, and the training set is not broken. The model-free curves say the same without any model: the median validation-to-training ratio of I goes from 0.63 to 0.84 and that of s from 0.67 to 0.91. On the heavy events alone the per-event factors are now identifiable (fake check 98 %), and the rain scale that the links ask for follows the radar-to-gauge ratio (rank correlation 0.64 over 18 events). No weather condition is linked to the antenna factor (0 of 6 tests); with 22 events this is descriptive.

The limits are stated plainly. The radar product is already adjusted with gauges, so this is not an independent test; g_e compares point gauges with path means; only 5 validation events carry a ratio; after the division the 5–10 mm/h validation cells drop out of the test, and they were among the worst; and the 2–5 mm/h validation cell at 28 GHz stays low.

9.6 The current WAA model (v3, provisional)

The model kept for the final step is the two-store model fitted on all link lengths, with the timing of the gauges:

$$
\begin{aligned}
A(t) &= W_f(t) + W_s(t) + L\,P_{\mathrm{band}}\big(R(t-d)\big) + c,\\
\frac{dW_f}{dt} &= \frac{\varphi\, a_{\mathrm{link}} F_{\mathrm{band}}(R) - W_f}{\tau_f}, \qquad \tau_f = 2.6 \text{ min while wetting, } 10.3 \text{ min while drying,}\\
\frac{dW_s}{dt} &= \frac{(1-\varphi)\, a_{\mathrm{link}} F_{\mathrm{band}}(R) - W_s}{\tau_s}, \qquad \tau_s = 240 \text{ min}, \quad \varphi = 0.72,
\end{aligned}
\tag{16}
$$

with a lag d of 1 minute and the level F at 28 GHz of 0.98, 1.63, 2.62 and 3.38 dB at 0.5, 2, 5 and 10 mm/h. Its known limits: the slow store sits at its bound and the attenuation after the rain is too high (1.2 to 1.4 times the observed integral), and its intervals come from short refits and are too narrow. This model was refitted on the corrected rain and tightened in S5a and S6 (Section 10); the WAA candidate of Section 10.3 replaces it.

In short. Treated as a system with rain as input, WAA has memory: it builds up in 2 to 3 minutes and dries in 9 to 13 minutes (gauge input, training and validation). A model-free test across link lengths shows an antenna part that saturates at 2 to 3 dB, so a constant WAA ($f = L + L_0$) does not describe the data. The models fitted on all lengths pass that test on training; the failure on validation is explained by the radar reading those events too high. The free rung M5 and the naming of an equation failed their checks.

10 Stage 1, part 5: the final model and a tighter equation (S5a, S6)

From here on the rain input is the radar divided by the event's radar-to-gauge ratio, $R = R_{\text{radar}}/g_e$, with g_e from the City gauges only (Section 9.5); $g_e = 1$ where the gauges caught too little rain. It is called the *corrected rain*. The raw radar is still reported next to it.

10.1 S5a: refit on the corrected rain, and one more slow store

The model of Eq. (16) was refitted on the corrected rain, on the training events of all link lengths (9780 events on a sub-link, 51 events), with intervals from 30 resampled sets of events, each fitted from a cold start. Four results:

- **The input closes the validation gap, not the refit.** On the corrected rain every model passes the length test on the validation events (5 of 6 cells), and on the raw radar every model fails it (at most 1 of 8), including the old fit that never saw the corrected rain. The refit moves the parameters little (F at 28 GHz and 5 mm/h: 2.71 against 2.62 dB).
- **Most of the slow leftover was the radar.** After the rain, the model's integral of A is now 0.79 times the observed one on links below 1 km and 1.21 times on links of 1 to 3 km, inside the accepted 0.7 to 1.4 range. In S2b to S3b it was 1.2 to 1.4. The median curve after the last shower is still too high from about 40 minutes on.
- **The shower peaks get worse.** Dividing the whole event by g_e also lowers the peaks, where the radar was not too high: the peak deficit grows from -0.53 to -1.05 dB. One ratio per event is the wrong correction at the peaks; a radar error that depends on the rain rate would explain it (an open question).
- **A more flexible slow store (M4c) is not identified.** M4c gives the slow store its own wetting and drying times. Its fake check passed, and by the rule written before the run it replaced M4b. On the real data, however, its wetting time ran to the 600-minute bound and its share φ is anywhere between 0.19 and 0.84; it gains only 0.0005 dB on the validation events. The rule had not asked for identified parameters; it should have.

First closed forms were also fitted: the path part came out as a straight line in R (a power just below 1 at 38 GHz), the antenna level as a saturating curve whose height is poorly determined.

A decision, stated openly. On 18 October 2026, before any test data were read, the final model was set to M4b with the gauge timing fixed (Eq. (16) on the corrected rain), and M4c was dropped. This deviates from the S5a rule. It was decided after seeing the validation events; the reason is that M4c's slow store is not identified, not its validation score. The planned step S5b (read the test set once) was not run; its parts moved to S6, S8 and S9.

Integrity rules from S6 on. Eight rules were written down at the same time: every rule and threshold is fixed before the run; a fake check runs before real numbers are trusted; a quantity is reported as a result only if it is identified (recovered within ±20 % on fake data, and its interval does not touch a bound); every deviation from a written rule is dated with its reason; failed checks are reported, never dropped (S7 may send the work back to S6 at most once); validation numbers are labelled optimistic; the test set is read once, after a frozen and hashed model, and nothing changes after it; and the standard power law of the ITU never corrects data.

10.2 S6: tightening the equation

S6 takes the final model on the corrected rain and asks how well each of its numbers is determined. Every rule was written before the run; no test data were read.

What the data are. Of the 728 sub-links, 660 are in the three frequency bands used (28, 32 and 38 GHz; the 7 to 23 GHz sub-links are dropped by the band rule, not by quality), 518 of them on training links. Of the 29 998 pairs of a training event and a training sub-link, 9780 remain (33 %). The largest loss is that the link saw no rain in the event (13 480 pairs); the quality rules remove about a fifth of the rest. The data are mostly dry: of 9.5 million link-event minutes, 12 % have at least 0.5 mm/h and only about 0.4 % have 10 mm/h or more. Everything said about F and P above 10 mm/h rests on those few minutes.

One shape for all bands. The free curves F and P have one value per band at each of seven rain rates (0.5 to 40 mm/h). S6 tested a tighter form: one shape shared by the bands and one amplitude per band,

$$F_{\text{band}}(R) = A_{\text{band}}\, G(R), \qquad P_{\text{band}}(R) = B_{\text{band}}\, H(R), \qquad G(5) = H(5) = 1. \tag{17}$$

The rule: keep it if the validation error rises by at most 0.003 dB and the length test still passes. Both shared shapes passed (0.3875 against 0.3878 dB, 5 of 6 cells) and were kept. On the training events an information criterion still prefers band-specific shapes, for F also after allowing for the correlation in time of the residuals, so the bands do differ in shape on training; the difference does not carry over to the validation events.

Honest intervals. The main interval comes from 60 resampled sets of whole events, each fitted from a cold start; all 60 converged. The amplitudes of the antenna level at 5 mm/h are $A = 2.74$ [2.20, 3.37], 2.56 [1.88, 3.39] and 2.96 [2.49, 3.53] dB at 28, 32 and 38 GHz. A second interval, from the profile likelihood, is up to 13 times narrower for the shapes and the path amplitudes: it counts every 5-minute point as independent once the correlation in time is allowed for, but all links of one event share the same rain error, so the independent pieces are closer to the 51 events. It is a lower bound only. For the antenna amplitudes the profile is *wider*, and that reveals a limit of the model: raising A_{band} and lowering every a_{link} of the band by the same factor changes nothing in the prediction. **The data determine the product $a_{\text{link}}F$, not F alone**; the level of F is that of a link with $a_{\text{link}} = 1$, set by the weak prior that keeps the median link amplitude near 1.

Timing. The wetting time (2.4 or 3.1 instead of 2.6 minutes) moves every number by less than 2 %. The drying time (9.2 or 11.8 instead of 10.3 minutes) moves F by up to 11 % and P by up to 19 % (most at 1 mm/h), in opposite directions: a longer drying keeps W up longer, so less is left for the antenna level and more for the path. Every change is inside the interval of its number.

Identifiability. Fake data were made from the fitted model on the real geometry, rain and noise, three times. All 62 quantities came back within ±20 % in all three (largest error 10 %). Eight are nevertheless *not identified*, because their real interval reaches zero: the path part below 1 mm/h. The fake data cannot test the trade-off between a_{link} and F (the truth obeys the same prior), and the slow store is fixed, so it is not tested either.

Closed forms: physics against symbolic regression. Formulas were fitted to the learned curves, with the uncertainty of each point from the resampling. Physics candidates (for F: a saturating hyperbola, a saturating exponential, a capped power; for P: a line, a power, a line with a square term) and the formulas proposed by symbolic regression (PySR [7]) competed under the same rule: the corrected Akaike criterion (AICc), and a candidate within 2 of the best with fewer parameters wins. The chosen form was then put back into the dynamic model and refitted (the *plug-back*); it must cost at most 0.005 dB on the validation events and keep the length test.

- **The path part is a straight line**, $P_{\mathrm{band}} = c_{\mathrm{band}}\, R$, with $c = 0.107$, 0.142 and 0.197 dB/km per mm/h at 28, 32 and 38 GHz. A power law gives an exponent of 0.97, and the square term is zero. The plug-back holds. This line was found among the candidates; the standard power law was not used.
- **The antenna level rises like $\sqrt{R}$**: the form chosen is PySR's $(R/5)^{0.47}$, within 0.2 of the capped power with one parameter fewer. The saturating hyperbola of S5a is 11 AICc worse with the corrected rain and the shared shape. Its plug-back, however, lowers the validation error a little but brings the length test down to 4 of 6 cells. That is exactly two thirds; the threshold written since S3b is 0.6667, so it *fails* by a hair, and the free shared curve of F is kept. Whether "at least two thirds" was meant is a decision left to the learners, to be dated (integrity rule 4).

Three frequencies are too few for a law in the frequency: the amplitudes describe these three bands only.

10.3 The WAA candidate

$$
\begin{aligned}
A(t) &= W_f(t) + W_s(t) + L\, c_{\mathrm{band}}\, R(t-d) + c, \qquad R = R_{\mathrm{radar}}/g_e, \quad d = 1 \text{ min},\\
\frac{dW_f}{dt} &= \frac{\varphi\, a_{\mathrm{link}} A_{\mathrm{band}} G(R) - W_f}{\tau_f}, \qquad \tau_f = 2.6 \text{ min while wetting, } 10.3 \text{ min while drying},\\
\frac{dW_s}{dt} &= \frac{(1-\varphi)\, a_{\mathrm{link}} A_{\mathrm{band}} G(R) - W_s}{\tau_s}, \qquad \tau_s = 240 \text{ min}, \quad \varphi = 0.72 \text{ (fixed, not identified)}.
\end{aligned}
\tag{18}
$$

Table 13 gives its numbers and Figure 17 the antenna level.

Table 13: The WAA candidate of S6. Brackets: 90 % intervals from 60 resampled sets of training events. G is the shared shape of the antenna level, linear between the points, zero at $R = 0$.

	28 GHz	32 GHz	38 GHz
A_{band} (dB)	2.74 [2.20, 3.37]	2.56 [1.88, 3.39]	2.96 [2.49, 3.53]
c_{band} (dB/km per mm/h)	0.107	0.142	0.197
G at 0.5, 1, 2, 5 mm/h: 0.35, 0.35, 0.63 [0.38, 0.80], 1 (by definition)			
G at 10, 20, 40 mm/h: 1.21 [1.01, 1.49], 1.73 [1.18, 2.30], 1.97 [1.63, 2.81]			
a_{link}, 10 to 90 % of the links: 0.76 to 1.23			
Not identified: the path part below 1 mm/h; φ and τ_s (fixed)			

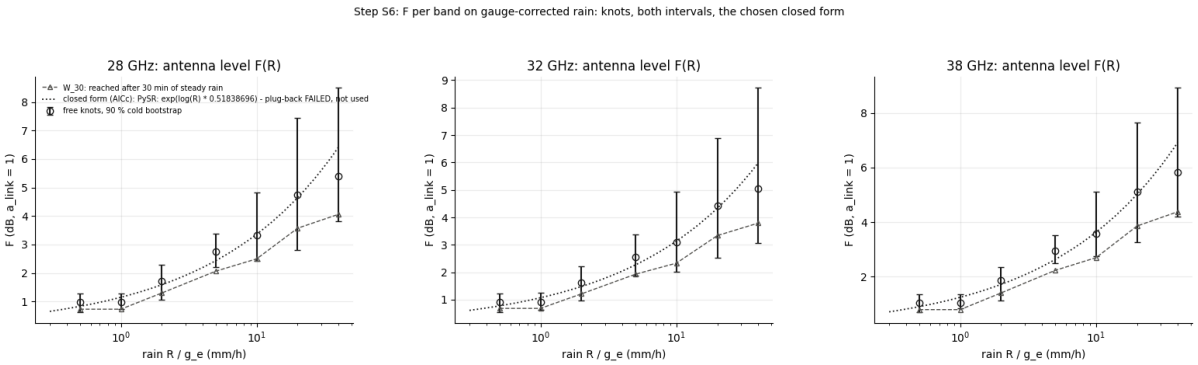


Figure 17: The antenna level F per band against the corrected rain (log scale), for a link with $a_{\mathrm{link}} = 1$. Circles with bars: the learned curve with 90 % intervals. Dashed line with triangles: the level W reached after 30 minutes of steady rain. Dotted line: the closed form $\propto R^{0.47}$, whose plug-back failed and which is not used.

The candidate rests on assumptions that S7 must check: that W does not depend on L (the level of F is tied to a_{link}, so a length effect could hide there), that the corrected rain is right at the scale of the path, that the gauge timing holds with the radar input, that one amplitude per link holds over the summer, and that dew and other wetting without rain are not part of it.

In short. On the corrected rain the final model is the two-store model with gauge timing; a more flexible slow store was dropped because it is not identified (a dated decision, made after seeing the validation events). S6 gives it one shape for all bands, intervals from 60 cold refits, and a fake check for every number. The path part is a straight line, $P = c\,R$. The antenna level rises about like $\sqrt{R}$, but that formula failed its check at exactly two thirds of the cells, so the learned curve is kept. The data fix $a_{\mathrm{link}}F$, not F; the path part below 1 mm/h and the slow store are not identified.

11 Where we stand

Table 14: What is established, and what is not, on 19 October 2026.

Established	**Not established yet**
$A(t)$ exists for every training and validation event on the training links, with a baseline that is unbiased and accurate to about one quantization step, and that does not absorb the drying.	Baselines for the test events and test links (needed for the final held-out check).
A drying law cannot be read from drying tails: tails wait or rise because of the event, and most of a tail passes while it still rains nearby (Section 8).	What makes whole events wait or rise after the rain (dew, drizzle under the radar, or no drying at night).
WAA has memory. With a gauge as input it builds up in 2 to 3 minutes and dries in 9 to 13 minutes, on training and on validation events.	One drying time for both inputs (with the radar the fitted drying is slower).
The slow wetting seen with the radar is the smoothing of the radar input, not the antenna (fake check).	The slow remainder after the rain: with the corrected rain most of it is gone, but after about 40 minutes the median is still above the model, and no identified store describes it.
The model-free antenna part saturates at 2 to 3 dB while the path part grows with the rain; a constant L_0 does not describe the data.	Whether everything in the intercept is the antenna (it holds anything that does not grow with L).
Fitted on all link lengths, the model reproduces the model-free test on training; the validation gap is explained by the radar-to-gauge ratio.	An independent check of that explanation (the radar is gauge-adjusted; 5 validation events).
The free rung M5 and SINDy-style naming do not give an identified equation on these data.	A readable equation beyond the store models; what the antenna factor of an event depends on.
A WAA candidate on the corrected rain (Eq. (18)): one shape for all bands, intervals from 60 cold refits, every number checked on fake data; the path part is a straight line $P = c\,R$.	A closed form for the antenna level (the $\sqrt{R}$ form failed its check at exactly two thirds); the level of F apart from a_{link}; the path part below 1 mm/h; the slow store.
The slow leftover after the rain was mostly the radar reading too high.	The shower peaks, which the corrected rain makes worse (-1.05 dB): a radar error that depends on the rain rate?
	Validation of the candidate without the test set (S7), and the test set itself (S8).

Two methodological habits have carried the work and will continue. The first is to *build a case with a known truth* before trusting a method: fake events for the baseline, simulated tails, and fake data driven by the real rain for every rung of the ladder and for every number of the final equation. In the system identification three fake checks failed (the path curve before its repair, the free rung, the per-event factors on light events), and each failure stopped a result from being read. The second is to *fix the decision rule before the run.* It also cuts the other way: twice a result looked convincing but did not pass the rule (the short links test, the per-event factors), and it was reported as undecided or failed; in S6 the closed form of the antenna level failed by exactly one cell and was not used. Where a written rule was overridden (M4c dropped in favour of M4b; a planned cut of the resampling from 60 to 40 sets undone), the decision was dated and its reason written down.

12 What comes next

12.1 The target of Stage 1

The general form of the goal is one dynamic equation for the WAA with rain as its input,

$$\frac{dW}{dt} = \underbrace{g(R, W)}_{\text{wetting}} - \underbrace{h(W)}_{\text{drying}}, \tag{19}$$

of which the store models of Section 9 are simple cases (a linear relaxation towards a level that depends on the rain, with different speeds for wetting and drying). The free version of this equation (M5) was not identified, so Stage 1 ends with the store model of Eq. (18), the WAA candidate of S6. Figure 18 shows the route.

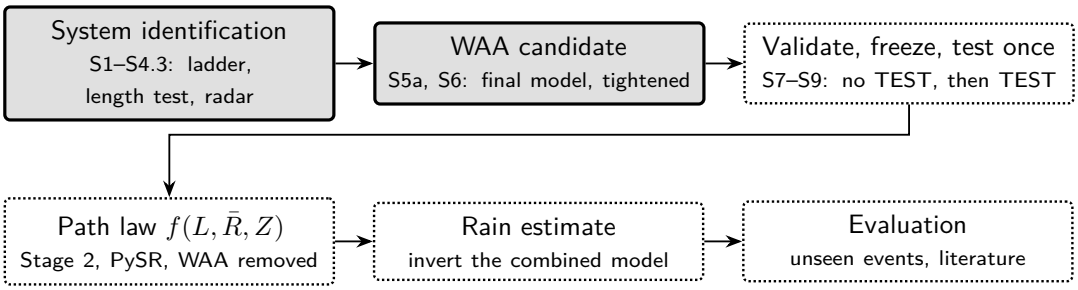


Figure 18: The remaining route. Grey fill: done. Dotted border: planned.

12.2 The steps

1. **S7, validate without the test set (not started).** Five checks of the candidate: that W does not depend on L (the link amplitudes and the W levels against the length); the short links of the validation events; the long links, at moments when the path is dry so that $A - c = W$, and pairs of links of different length on the same path, where $W = (L_2 A_1 - L_1 A_2)/(L_2 - L_1)$; dew (the same water layer?); and what is left over against the link metadata and the weather, with PySR, where a term enters only if it passes a fake and a held-out check. S7 may send the work back to S6 at most once.
2. **S8, freeze and test once (not started).** The model is frozen with a hash, and the test events and links are read once; nothing changes after that.
3. **S9, the write-up (not started).** The story, the assumptions with their checks, the data funnel and the core figures. The comparison with the WAA models of the literature [1], [9] belongs to the final write-up, with their values read from the papers.
4. **Publishing the fit caches.** When Stage 1 is finished, the saved fits are published so that others can rebuild the reports without hours of fitting.

12.3 After Stage 1

- **Stage 2, the path law.** With the WAA removed, learn $f(L, \bar{R}, Z)$ of Eq. (3) by symbolic regression (PySR [7]), with the radar rain along the path as the reference, and see how the dependence on the length differs from $f = L$. The model-free slope $s(R)$ of Table 11 is a first look at the path part.

- **Rain estimate and evaluation.** Invert the combined model (path law and WAA) to get rain from attenuation, on events that were not used for training. Measures: normalised bias and normalised RMSE against gauges and radar, by link length; the amount of false rain after the end of events; comparison with the plain power law, the WAA models of the literature [1], [9] and a neural-network reference [11]; and the stability of the learned terms under resampling.

12.4 Open questions

- What drives the slow remainder after the rain? With the corrected rain most of it is gone, but the median curve is still too high after about 40 minutes, and a slow store with its own times (M4c) is not identified.
- Why the radar and the gauge give different drying times on the same points.
- Is the validation gap really the radar? The radar product is gauge-adjusted, and only 5 validation events carry a radar-to-gauge ratio.
- The length-test threshold: 0.6667 as written, or "at least two thirds"? With two thirds, the $\sqrt{R}$ form of the antenna level would pass its plug-back. To be decided and dated.
- The level of F is set by the prior on the link amplitudes; only $a_{\text{link}}F$ is determined by the data.
- The shower peaks: one ratio per event lowers them too much; the radar error may depend on the rain rate (the 28 GHz, 2–5 mm/h validation cell also stays low).
- Does the antenna level saturate or keep rising slowly? S5a chose a saturating curve, S6 a $\sqrt{R}$ rise.
- The bands differ in shape on training but not on validation: a real band effect hidden by the shared shape?
- Warm events are over-predicted more (residual $-0.15\,\text{dB}$ against $-0.02\,\text{dB}$ in the cool ones): antenna, rain type or radar?
- Dew and night-time rises of the level, without rain.
- Uneven rain on long links may lower the slope s (the radar path mean on a 2 km grid); 32 GHz has no link below 1 km, so its intercepts are extrapolations.
- A free WAA law (M5) can only be named with conditions that keep W bounded (drying that rises with W, wetting that does not grow faster).
- The judge depends on the iteration cap of the fits (a few thousandths of a dB, the size of the differences between rungs).

References

The numbers are those of the project proposal; only the references used in this report are listed.

[1] Schleiss, M., Rieckermann, J., & Berne, A. (2013). Quantification and modeling of wet-antenna attenuation for commercial microwave links. *IEEE Geoscience and Remote Sensing Letters*, 10(5), 1195–1199.

[6] Brunton, S. L., Proctor, J. L., & Kutz, J. N. (2016). Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15), 3932–3937.

[7] Cranmer, M. (2023). Interpretable machine learning for science with PySR and SymbolicRegression.jl. arXiv:2305.01582.

[8] Ostrometzky, J., & Messer, H. (2018). Dynamic determination of the baseline level in microwave links for rain monitoring from minimum attenuation values. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.

[9] Pastorek, J., Fencl, M., Rieckermann, J., & Bareš, V. (2021). Precipitation estimates from commercial microwave links: practical approaches to wet-antenna correction. *IEEE Transactions on Geoscience and Remote Sensing*.

[10] Messenger, D. A., & Bortz, D. M. (2021). Weak SINDy: Galerkin-based data-driven model selection. *Multiscale Modeling & Simulation*, 19(3), 1474–1497.

[11] Habi, H. V., & Messer, H. (2020). Recurrent neural network for rain estimation using commercial microwave links. *IEEE Transactions on Geoscience and Remote Sensing*.

[13] Andersson, J. C. M., Olsson, J., van de Beek, R., & Hansryd, J. (2022). OpenMRG: open data from microwave links, radar, and gauges for rainfall quantification in Gothenburg, Sweden. *Earth System Science Data*, 14, 5411–5426.

A Glossary

Term	Meaning
$A(t)$	Attenuation: total loss minus baseline, Eq. (1).
A_{rain}, $\delta(t)$	The part of A caused by rain along the path; everything else.
$W(t)$	Wet-antenna attenuation (WAA), in dB.
$\bar{R}$, L	Mean rain rate along the path; path length.
$f(L, \bar{R}, Z)$	The path-law correction to be learned in Stage 2; uniform rain gives $f = L$.
Sub-link	One direction of a two-way link.
Baseline	The total loss of a sub-link without rain; here one number per (event, sub-link).
Fake event	A pretended event placed in certified dry weather, where the true attenuation is zero.
Offset, σ_{off}	The baseline error of one (fake) event; its spread for a sub-link.
Drying tail	The stretch of $A(t)$ after the rain has stopped on a link.
t_0, t_{trace}	End of the link's last wet bin; end of the last bin with any trace of rain (the tail start).
A_{start}	Mean of A over the first 5 minutes of the tail.
Control tails	Tails that start below 0.5 dB: noise without drying.
Tail set v1	The frozen set of 429 training tails (123 validation).
m1 ... m6, F1 ... F6	A drying law used as the truth of a simulation; the same law used as a candidate in a fit.
Spread s	Width of the tail-to-tail variation of the drying speed, Eq. (8).
BIC	Bayesian information criterion: goodness of fit with a charge per parameter; lower is better.
Pooled fit	One law with shared parameters fitted to all tails at once, Eq. (11).
Event bootstrap	Resampling whole events with replacement, to measure uncertainty.
SINDy	Sparse identification of nonlinear dynamics: finds a short equation for dW/dt from a library of terms.
Free curve $k(W)$	The drying rate per dB of water, drawn by the pooled fit without assuming a law, Eq. (13).
Jump 1, jump 2	Drawing the free curve from all tails; naming its equation by sparse regression.
Mixing trap	Exponential tails of different speeds, analysed together, look like a nonlinear law.

Term	Meaning
T0–T18, S1–S9	The steps of the drying-tail exploration (closed); the steps of the system identification.
Event on a sub-link	One rain event on one direction of a link: the unit of the system identification.
P, F, a_{link}	Path curve per km; antenna level curve; amplitude of one link, Eq. (14).
M0 . . . M5, M4b	The rungs of the model ladder (Section 9).
Stop rule	The simplest model whose difference to the best has an interval that holds zero.
Fake check	Data made from a known truth, which a method must give back before it is used on real data.
I, s, L_0	Model-free intercept (antenna part) and slope (path part per km) of Eq. (15); $L_0 = I/s$.
g_e	Radar rain divided by City-gauge rain for one event.
WAA v2, v3	The one-store model of S2b; the two-store model with gauge timing fitted on all lengths, Eq. (16).
Corrected rain	The radar rain divided by the event's g_e (from the City gauges only).
M4c	M4b with separate wetting and drying times for the slow store; not identified, dropped.
Shared shape	$F = A_{\text{band}}G(R)$, $P = B_{\text{band}}H(R)$: one shape for all bands, Eq. (17).
Cold refit	A fit of a resampled set of events from the same generic start, not from the answer.
Profile likelihood	An interval from how fast the fit gets worse when one number is held away from its best value.
AICc	Corrected Akaike criterion: fit quality with a charge per parameter, for small samples; lower is better.
Plug-back	Putting a closed form back into the dynamic model, refitting, and checking it on the validation events.
WAA candidate	The equation of S6, Eq. (18), before validation and the test set.

B Where the details are

This report gives the ideas. The exact rules, thresholds and numbers are in the documents of each module, inside the research folder.

Topic	Document
Baseline method, flags, accuracy, decisions	`baseline/METHOD.md`
Baseline validation	`R1_validation.md` and `R1b_validation.md`, in the reports folder of `baseline`
Tail definitions, rules of set v1, what each WAA step decided	`wet_antenna/METHOD.md`
Frozen tail set	`freeze_tail_set.md`, in the reports folder of `wet_antenna`
Simulator and its calibration	`calibrate_simulator.md`, same folder
Pooled fit on simulated tails	`check_pooled_fit.md`, same folder
SINDy on simulated tails	`check_sindy.md`, same folder
The two jumps and their confirmation	`check_free_curve.md` and `confirm_free_curve.md`, same folder
The rest of the method; the false step	`check_discovery_method.md` and `check_quantization.md`, same folder
The first fit of the real tails	`fit_real_drying.md`, same folder
Stalled tails; what is in a tail; drying after nearby rain	`diagnose_stalled_tails.md`, `characterise_tails.md`, `check_drying_after_rain.md`, same folder
Very short links	`short_links.md`, same folder
The model ladder (S1)	`system_id_ladder.md`, same folder
Links up to 3 km and the repairs (S2, S2b)	`system_id_s2.md` and `system_id_s2b.md`, same folder
Verification and the all-length fits (S3, S3b)	`system_id_s3.md` and `system_id_s3b.md`, same folder
Why events differ (S4.1–S4.3)	`conditions_s4.md` and `conditions_s4c.md`, same folder
Refit on corrected rain, M4c, first closed forms (S5a)	`final_s5a.md`, same folder
Tightening the equation; the WAA candidate (S6)	`tighten_s6.md`, same folder; the candidate as numbers in `tighten_s6_candidate.json`, in the products folder
Integrity rules and the decision on the final model	`docs/ROADMAP.md`, part B
The plan, and how to rerun every step	`docs/ROADMAP.md` and `docs/REPRODUCE.md`
The data layer	`docs/pipeline_log.pdf` (next to this report)